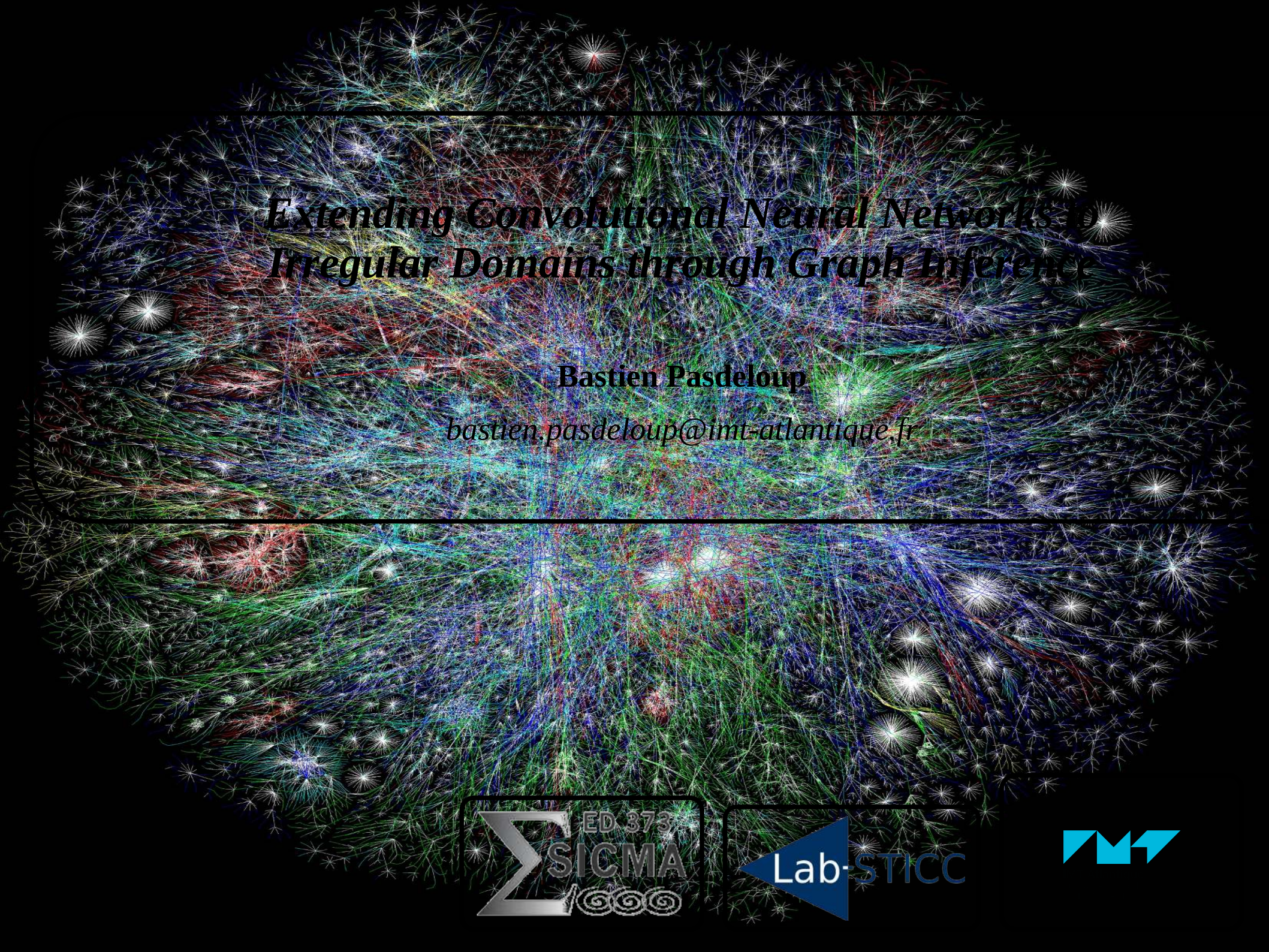




Extending Convolutional Neural Networks to Irregular Domains through Graph Inference

Bastien Padeloup

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Topics on graph signal processing

- * What is it?
- * The main ideas
- * Some problems and applications

Graph inference from signals

- * Why it may be necessary
- * The case of stationary signals

Translations on irregular graphs

- * How we define them
- * How we identify them
- * The theory and applications



Topics on graph signal processing

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- * Some problems and applications

- ▶ Graph inference from signals

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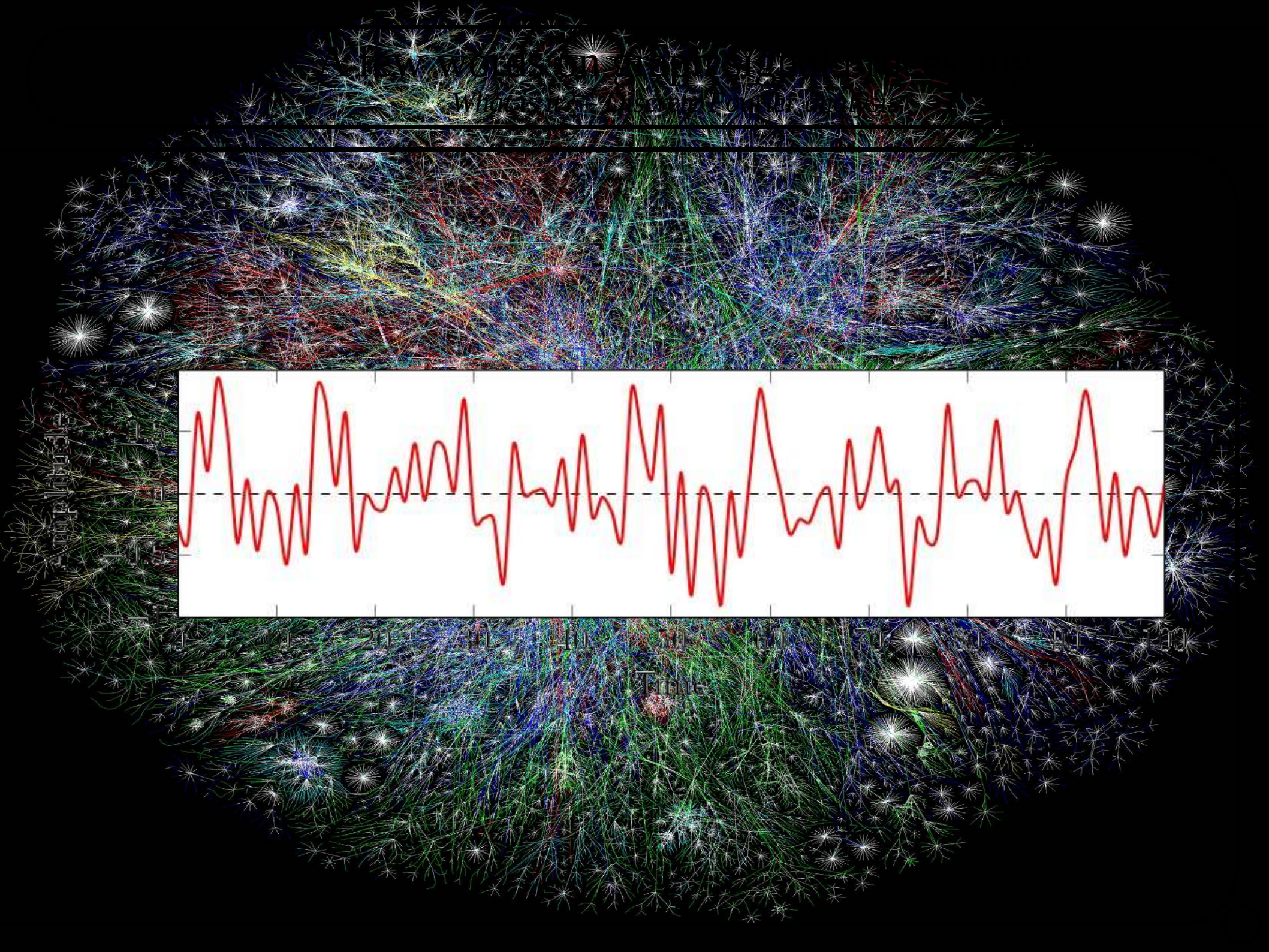
- * The case of stationary signals

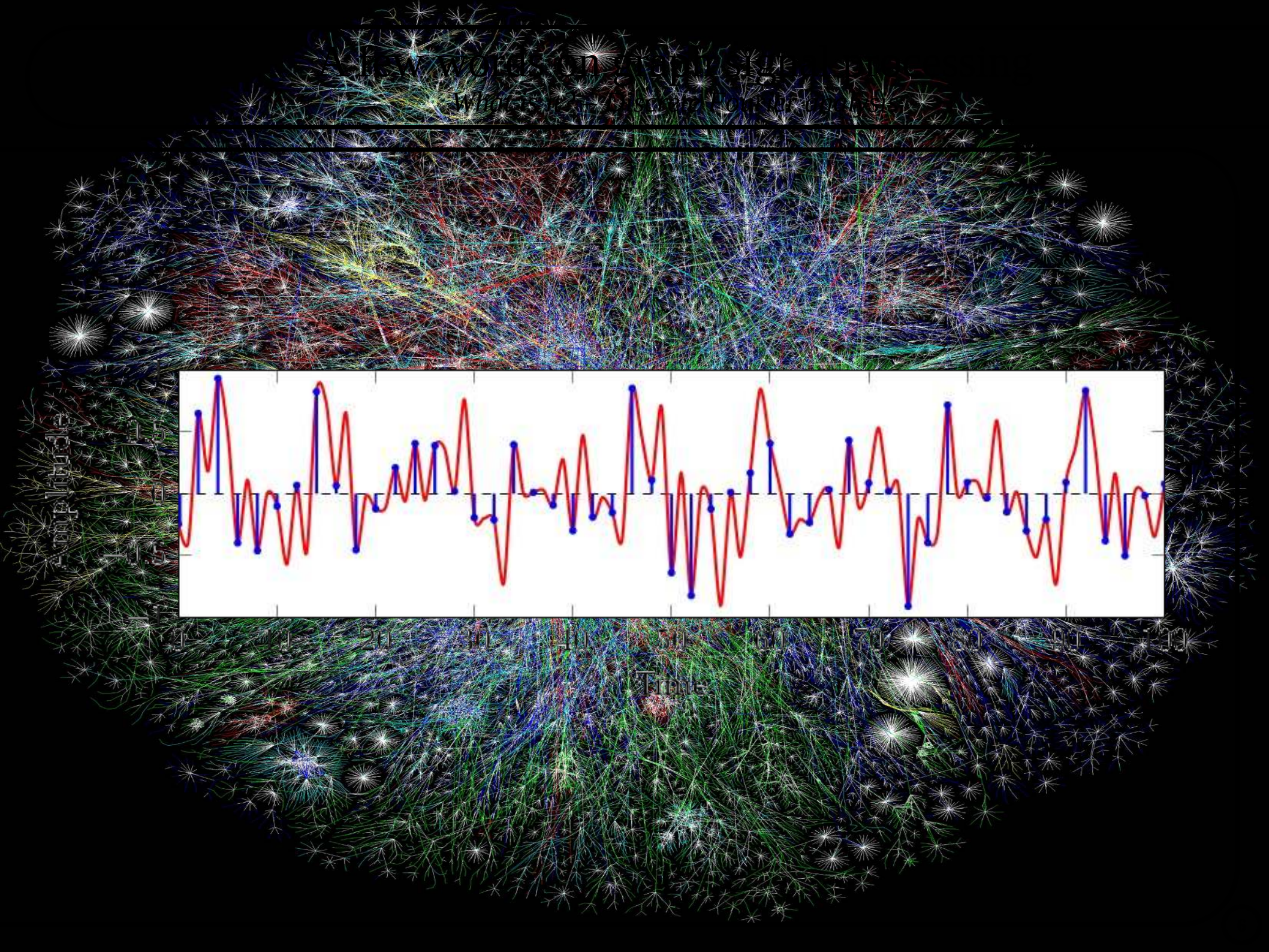
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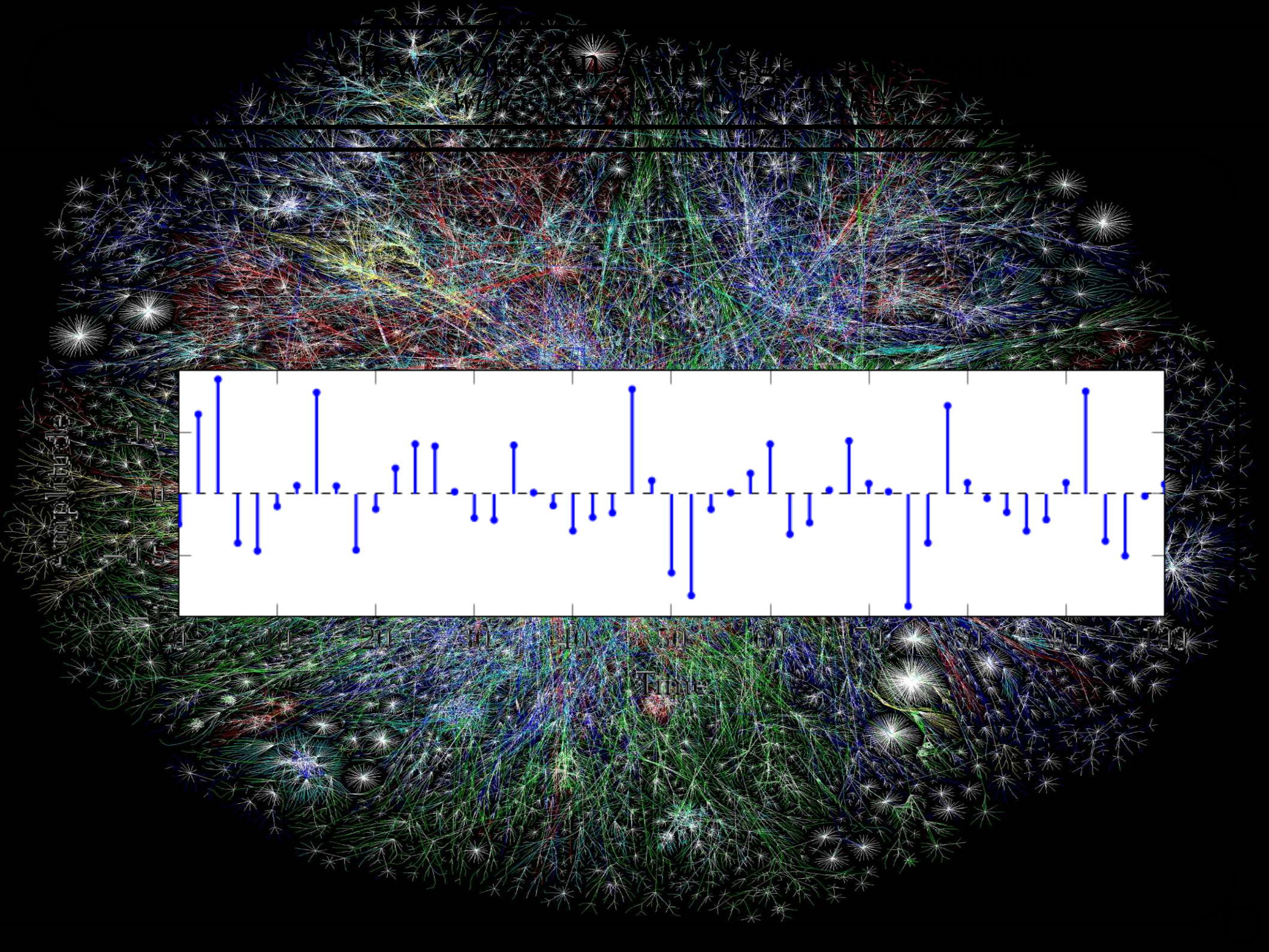
- * How we define them

- * How we identify them

- * Examples and applications

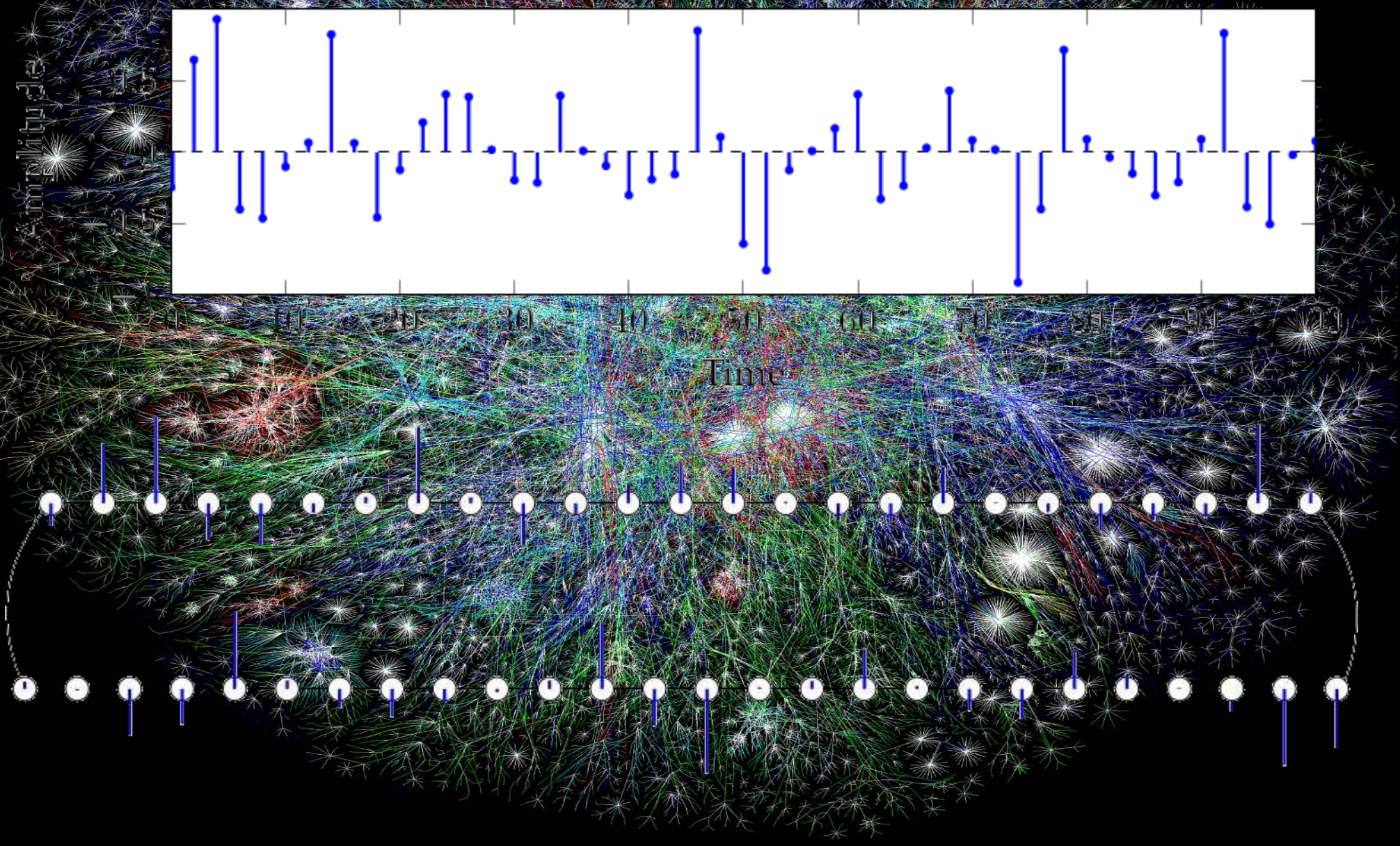




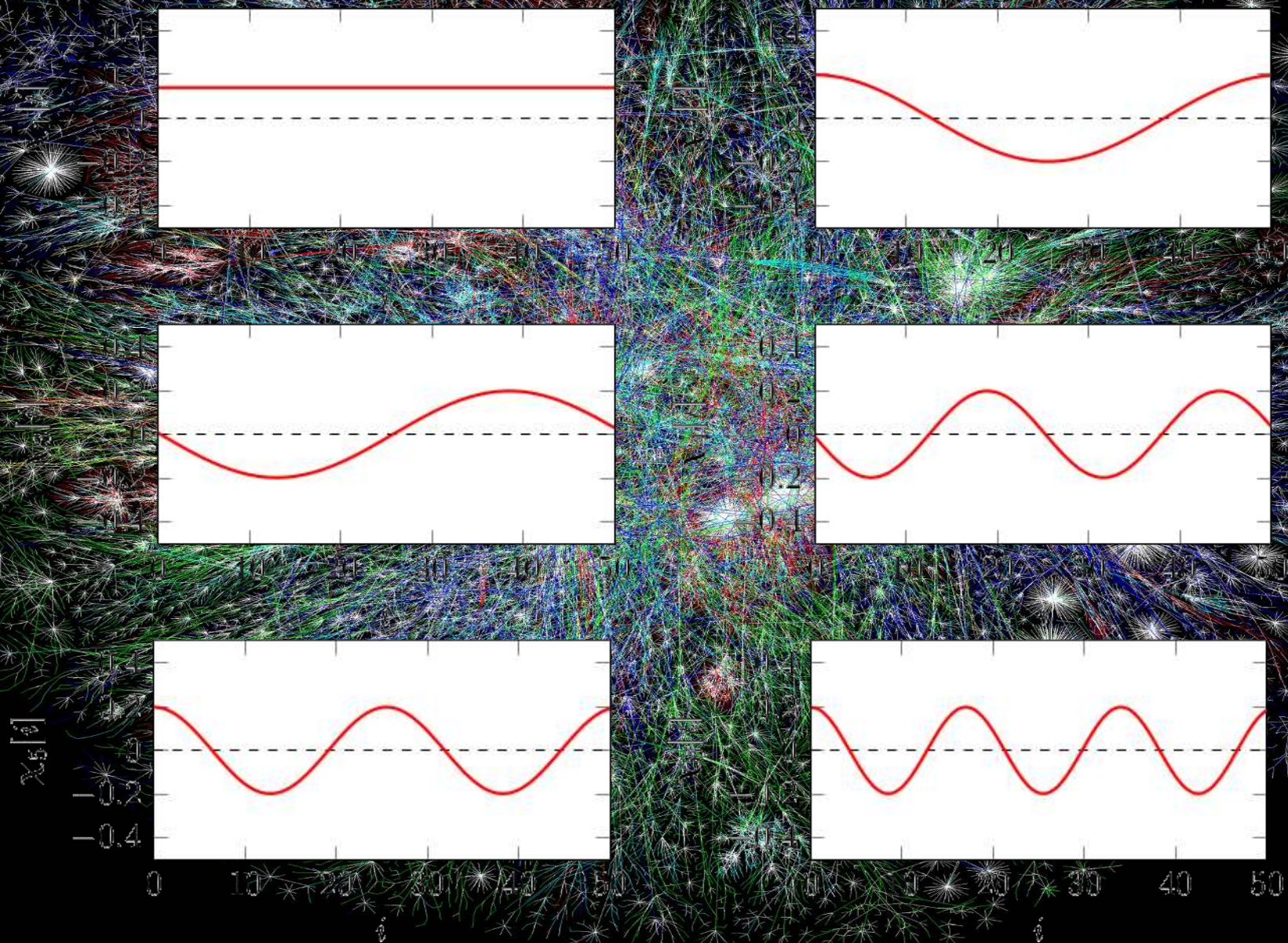


VIEW WORKS 10

What is it? A 100% natural product

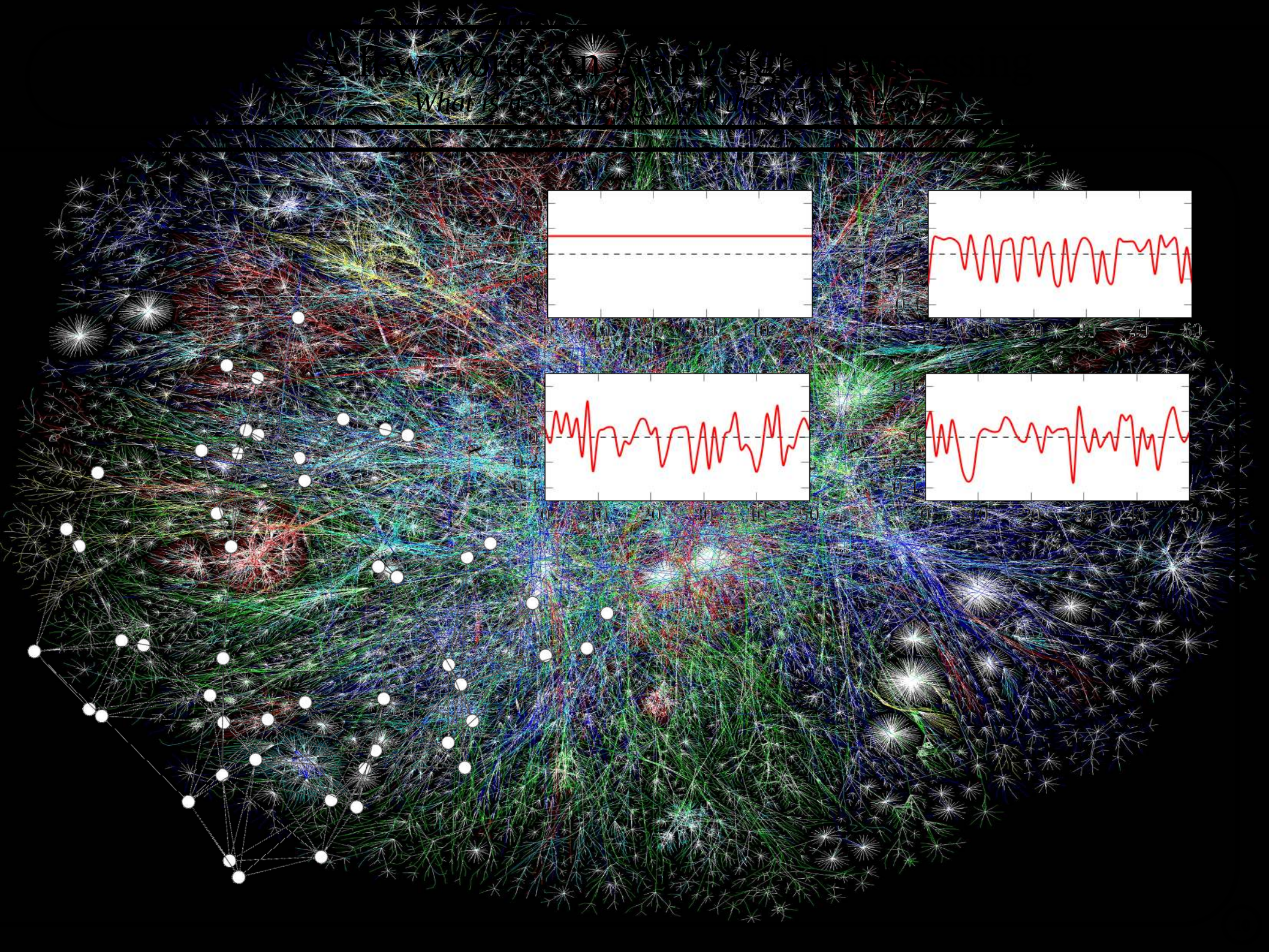


χ^2 vs θ — The first eigenmode of the linear fit



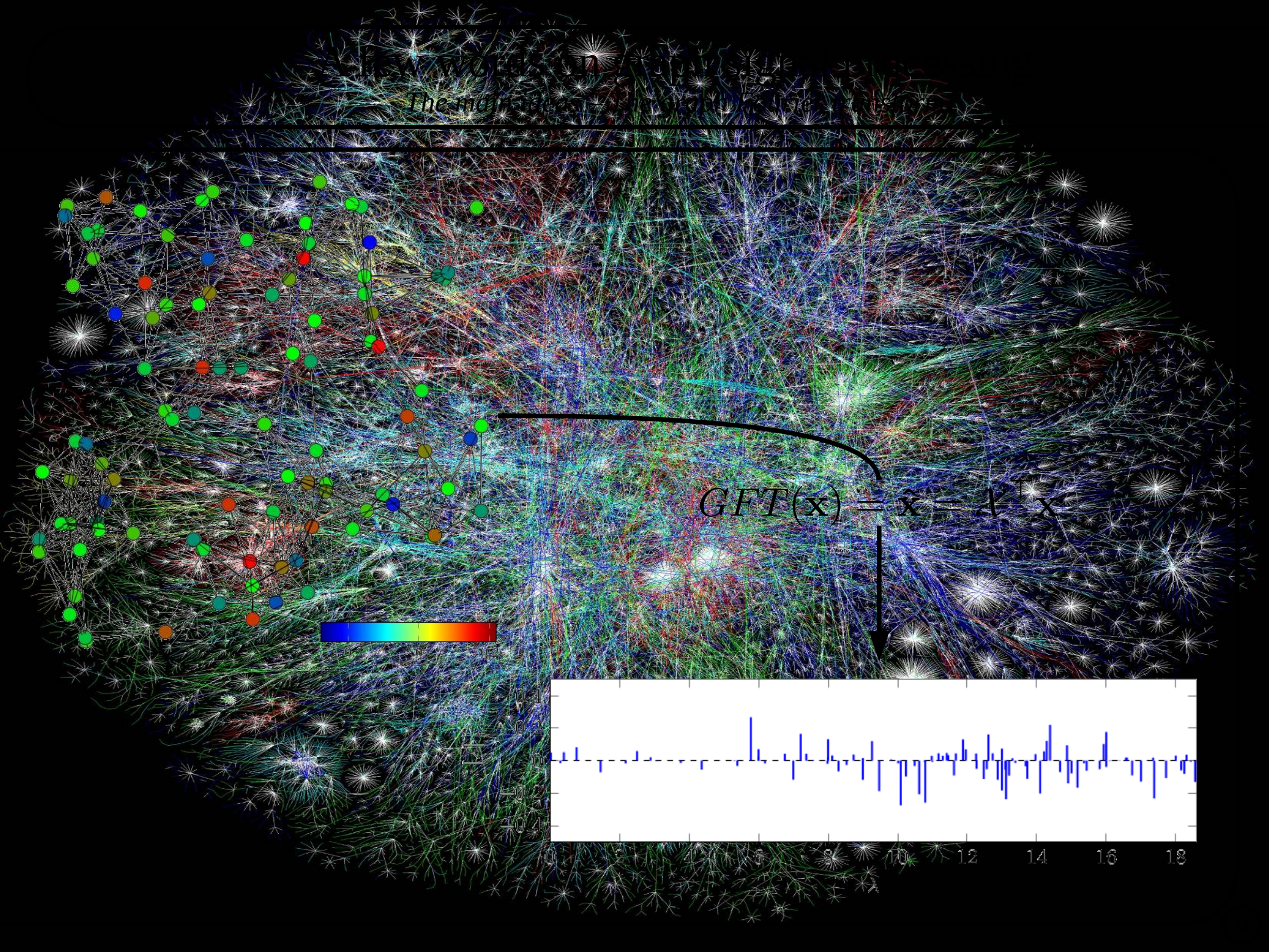
What's new in network science?

What is a network?

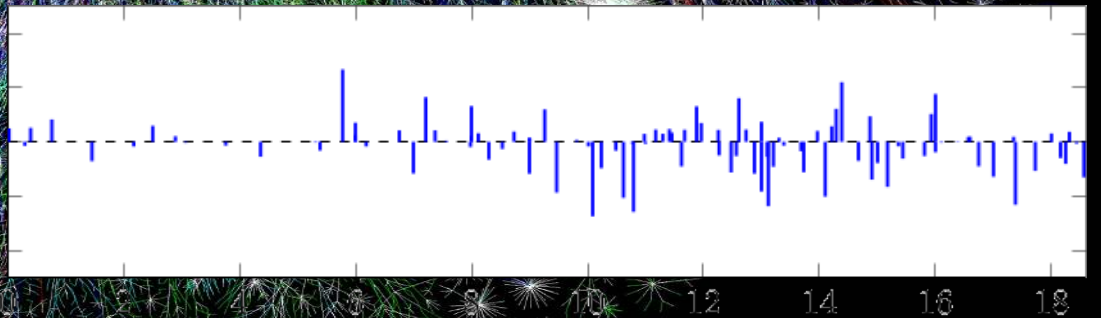


VIEW WORKS ON

The many faces of network science

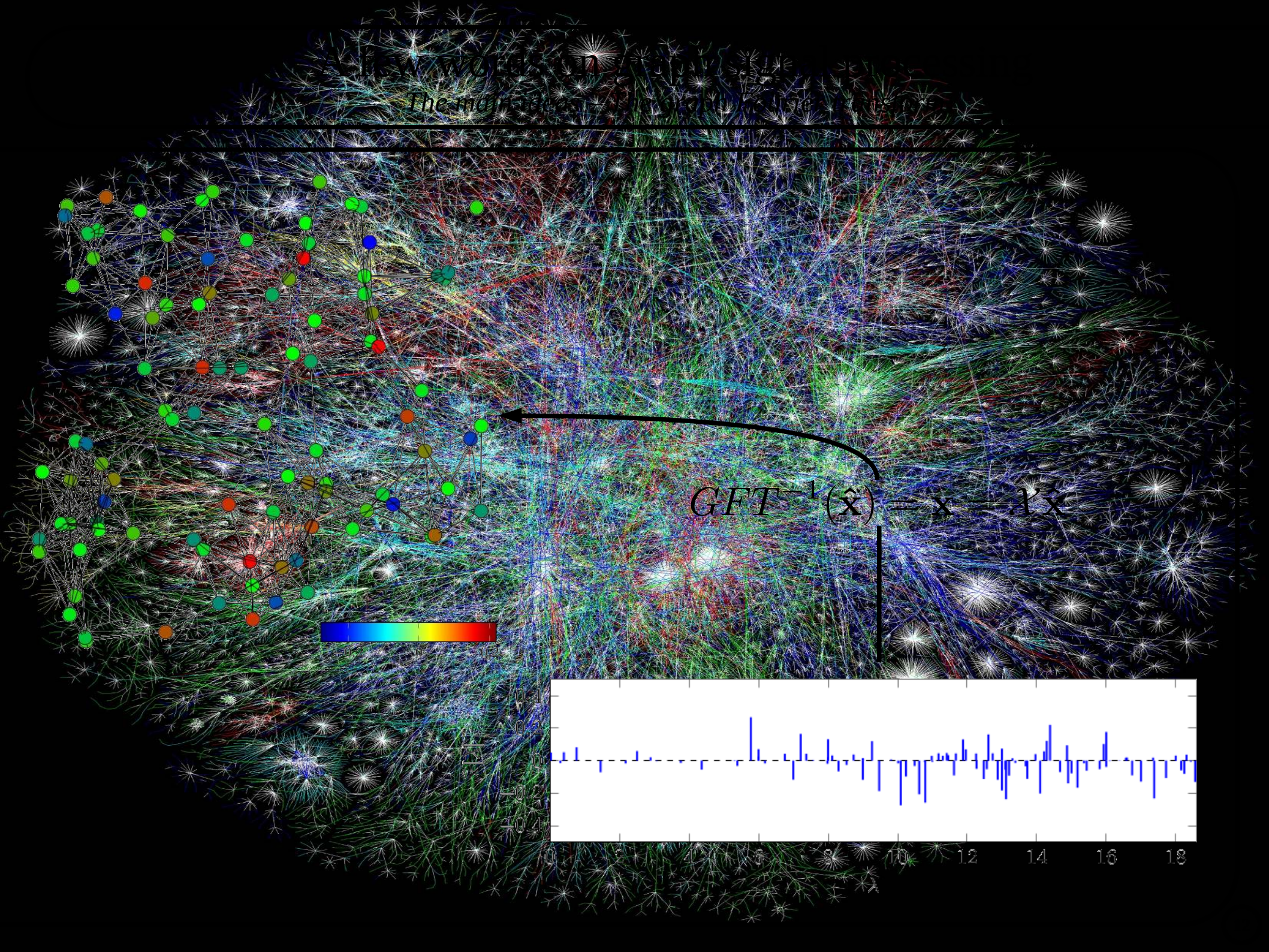


$$GFT(\mathbf{x}) = \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{x}$$

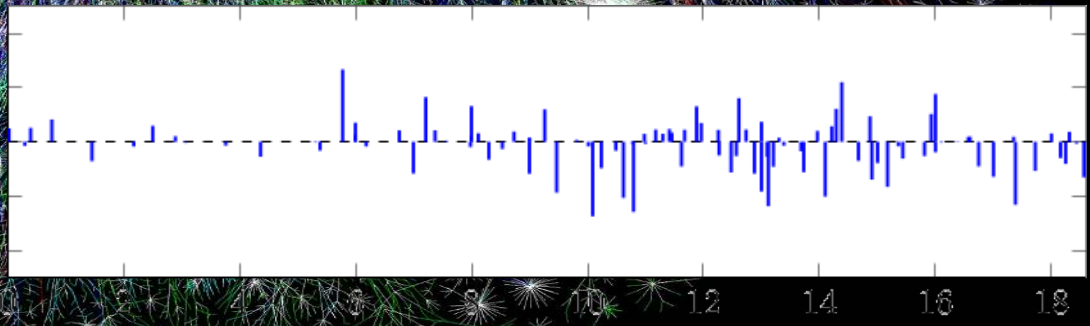


VIEW WORKS ON

The mean-field limit of the Kuramoto model

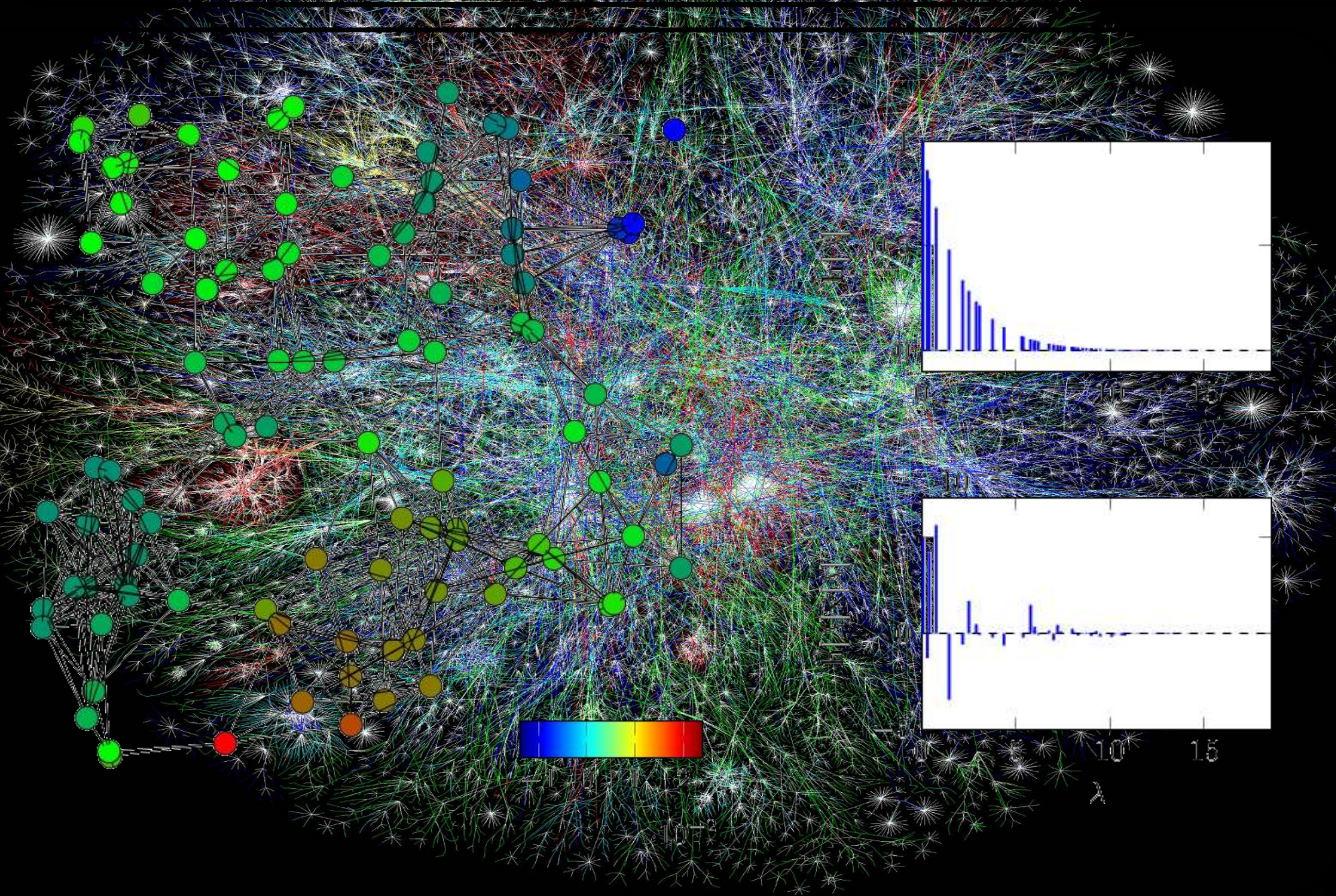


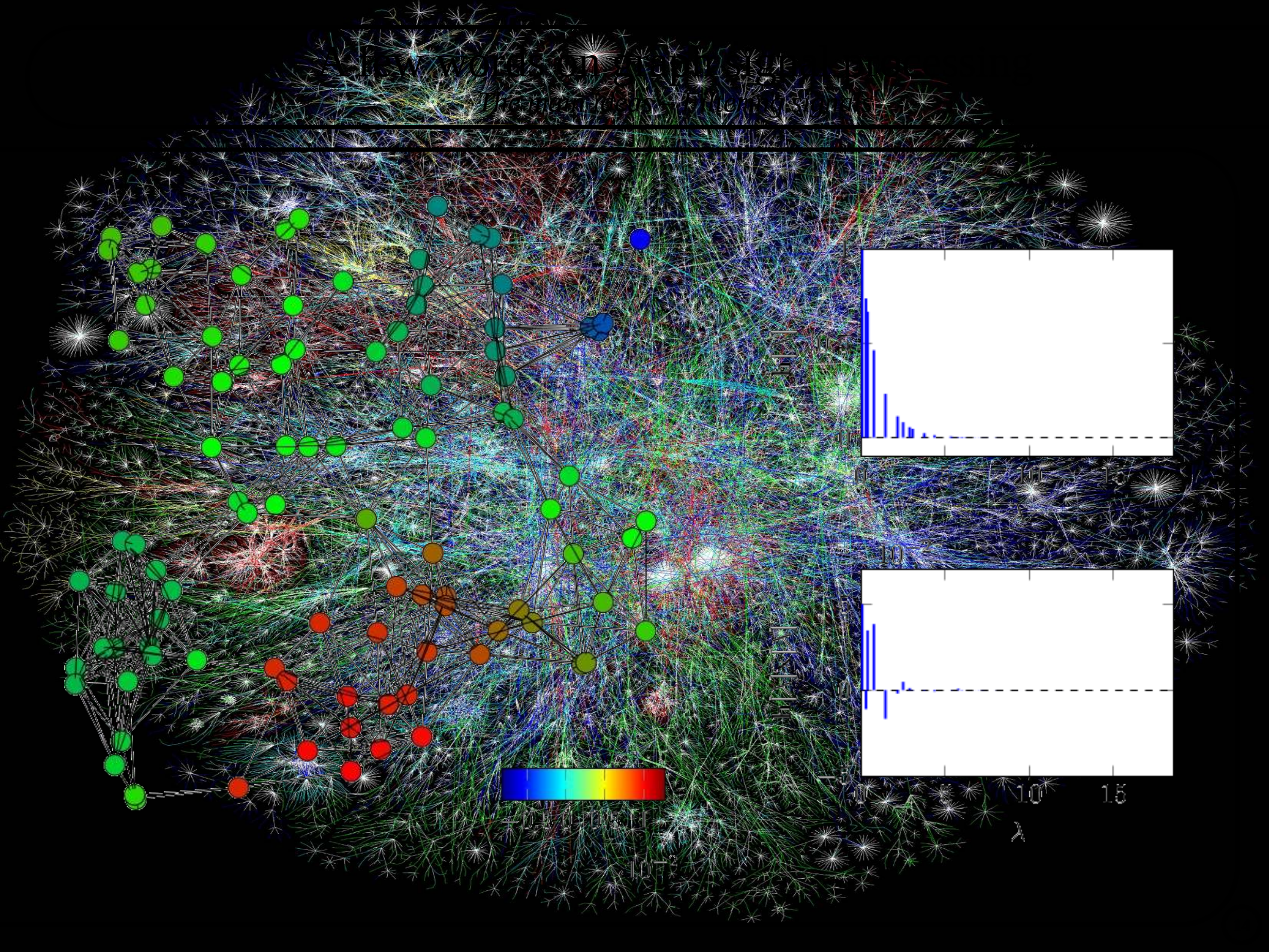
$$GFT^{-1}(\hat{x}) = x - \lambda \hat{x}$$



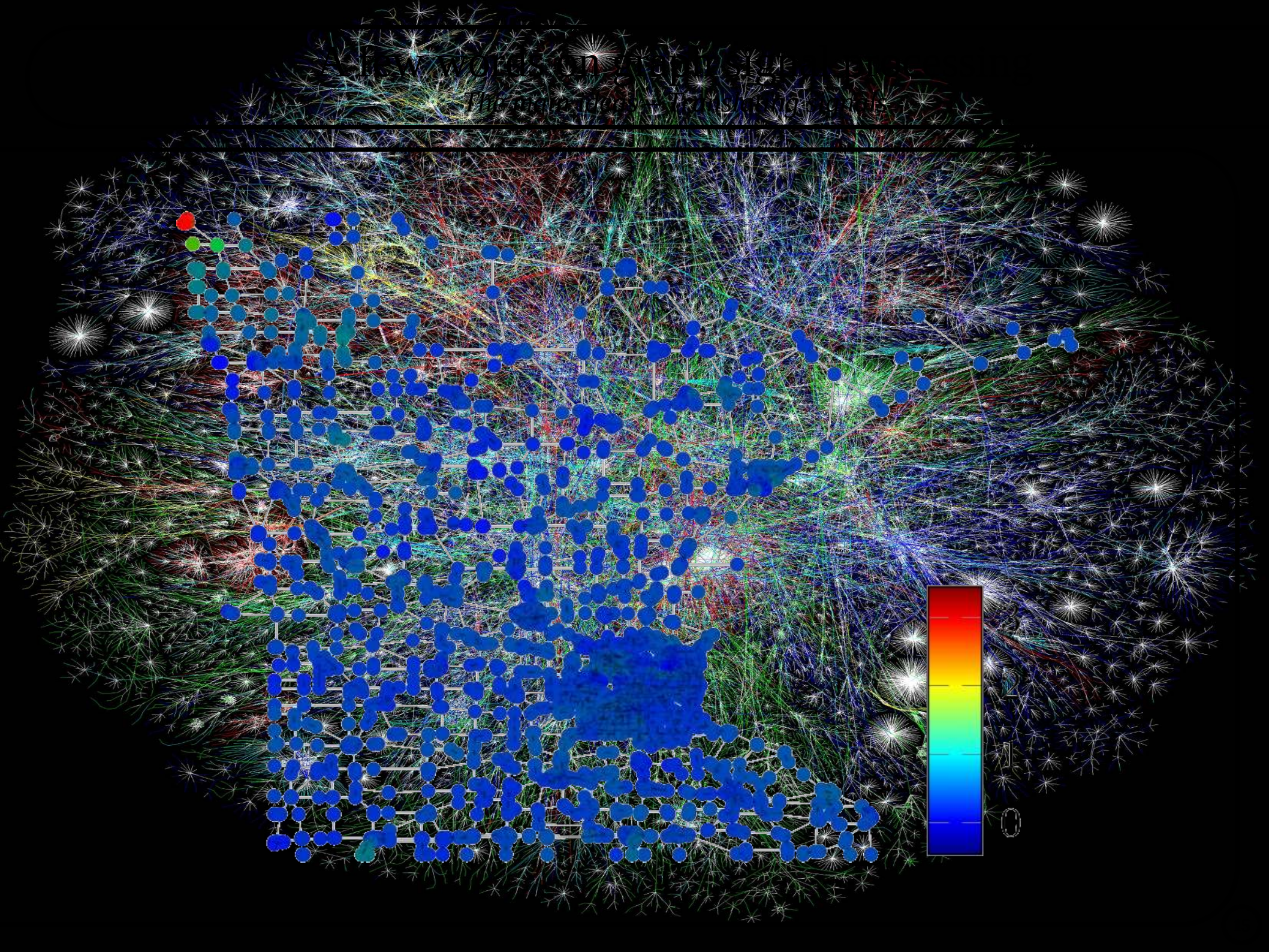
STANFORD UNIVERSITY

Department of Statistics

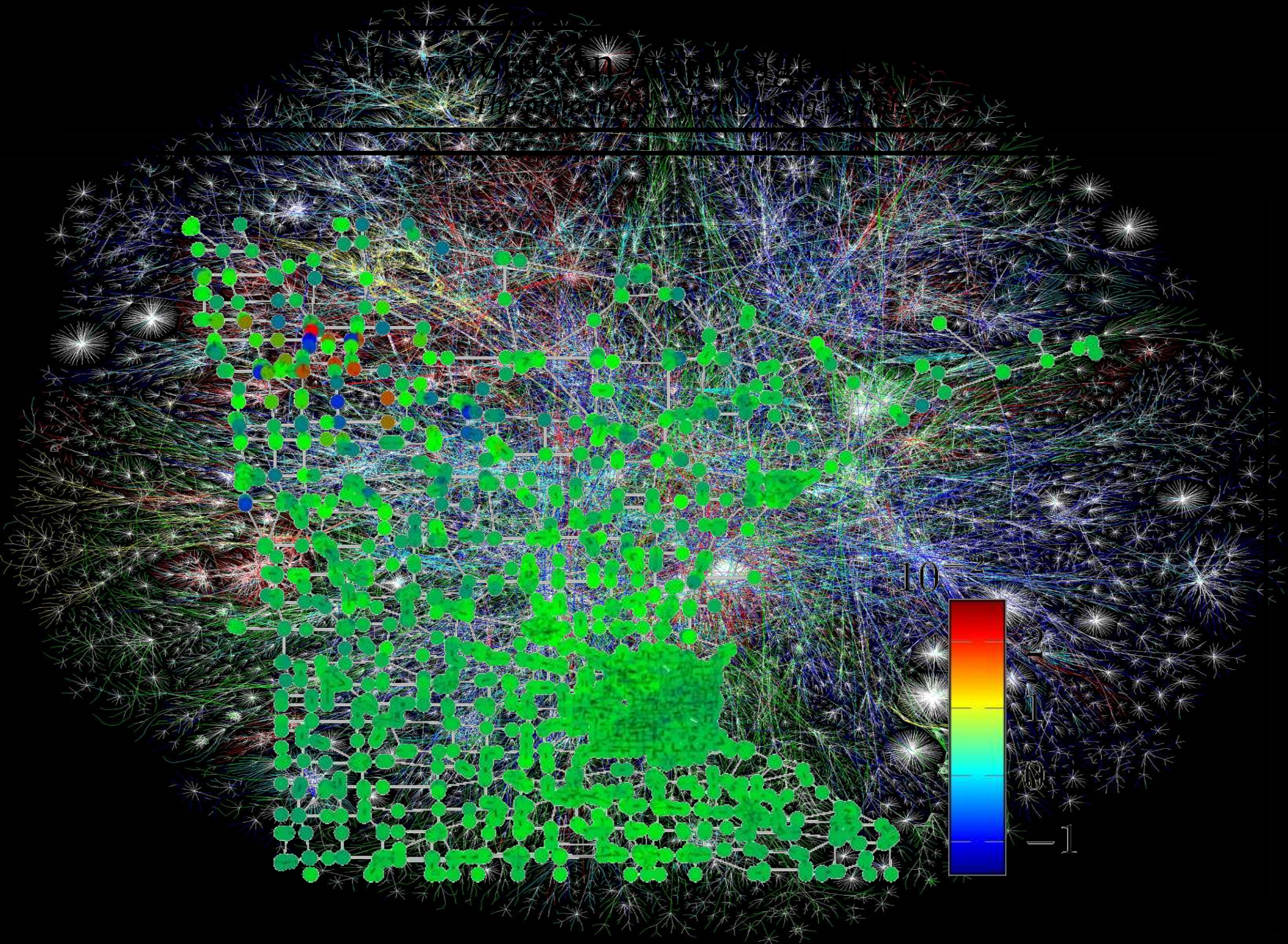




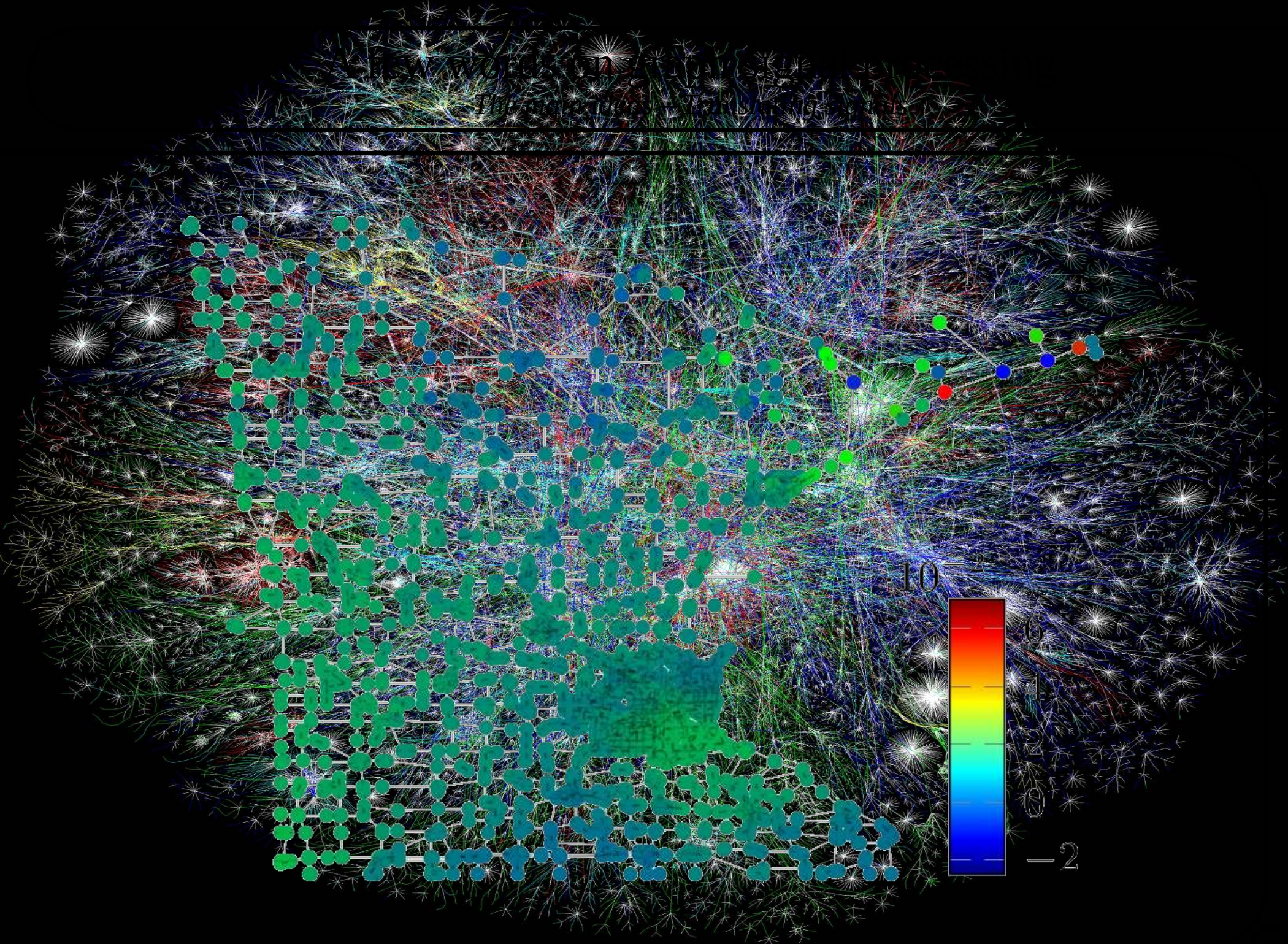
NEW WORK ON *The Geometry of the Universe*



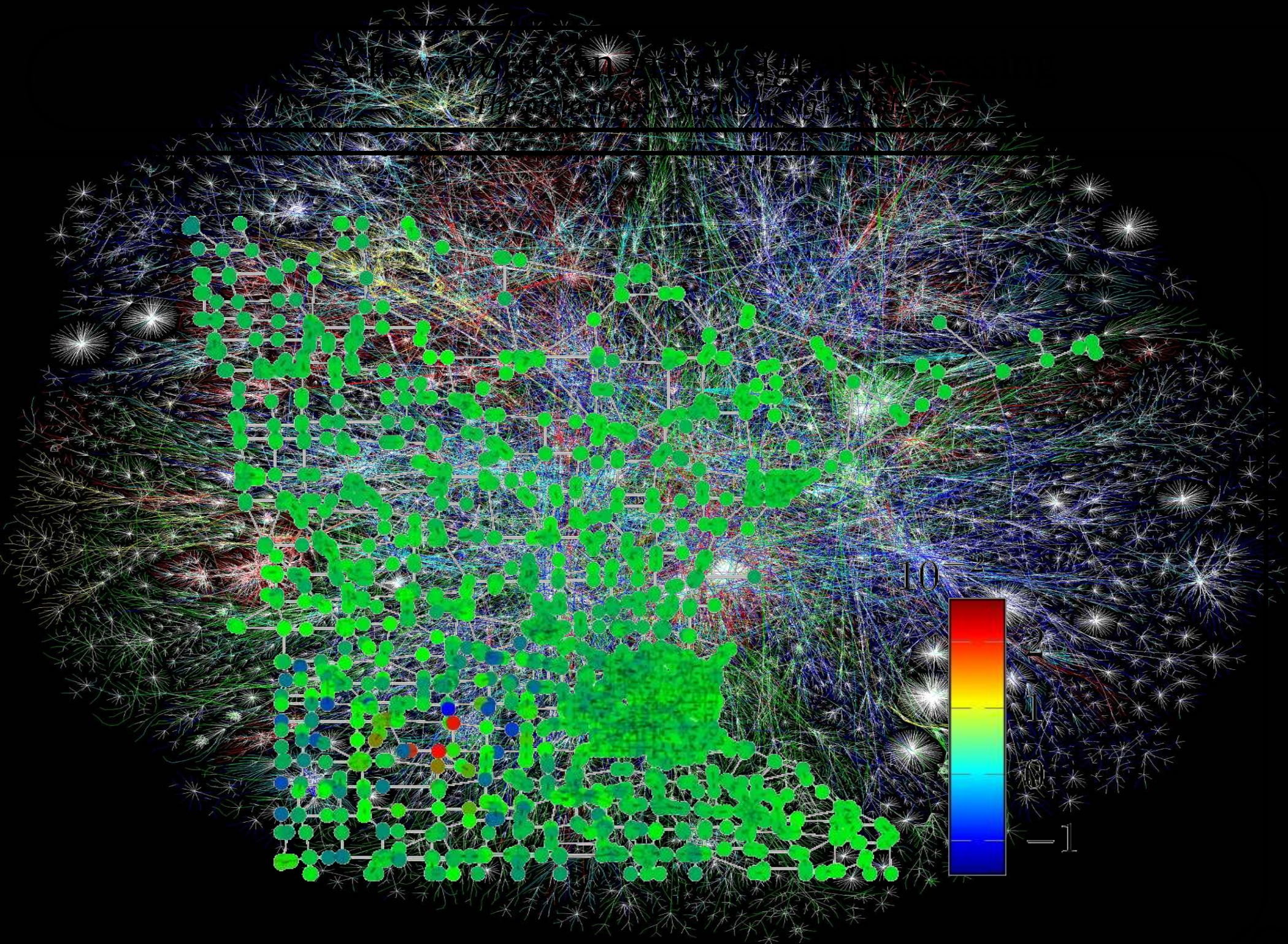
NEW WORK ON *The structure of the universe*

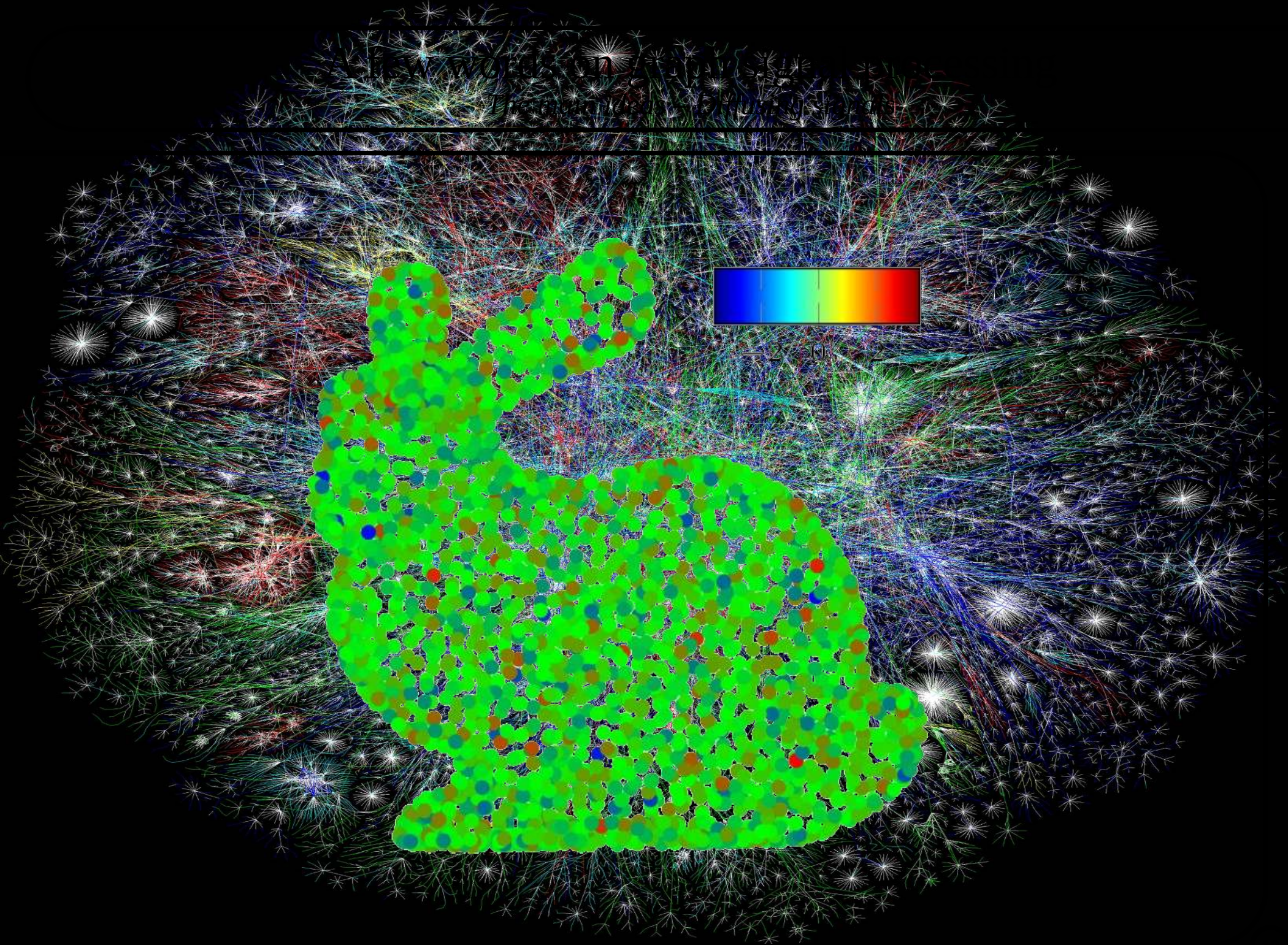


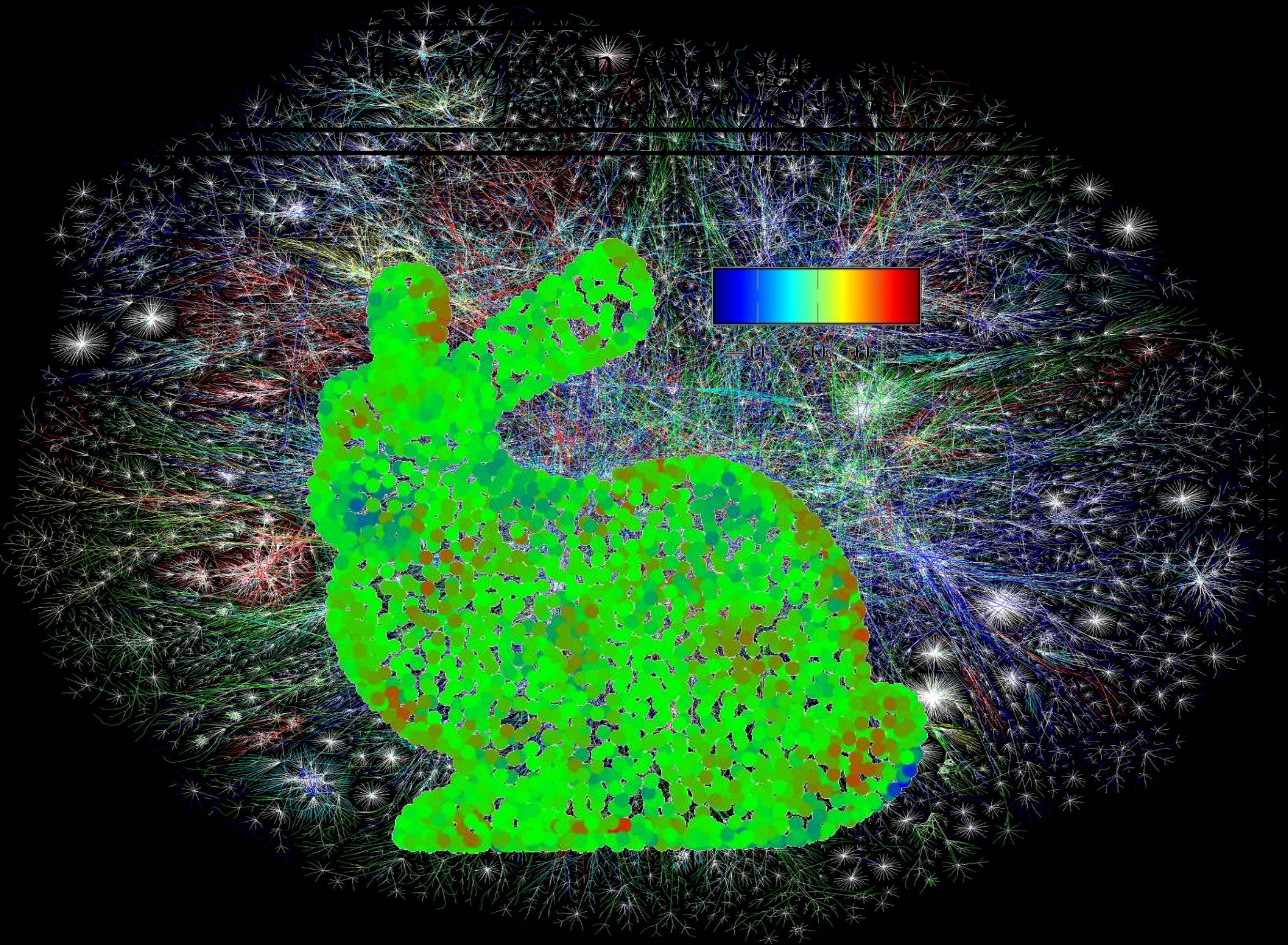
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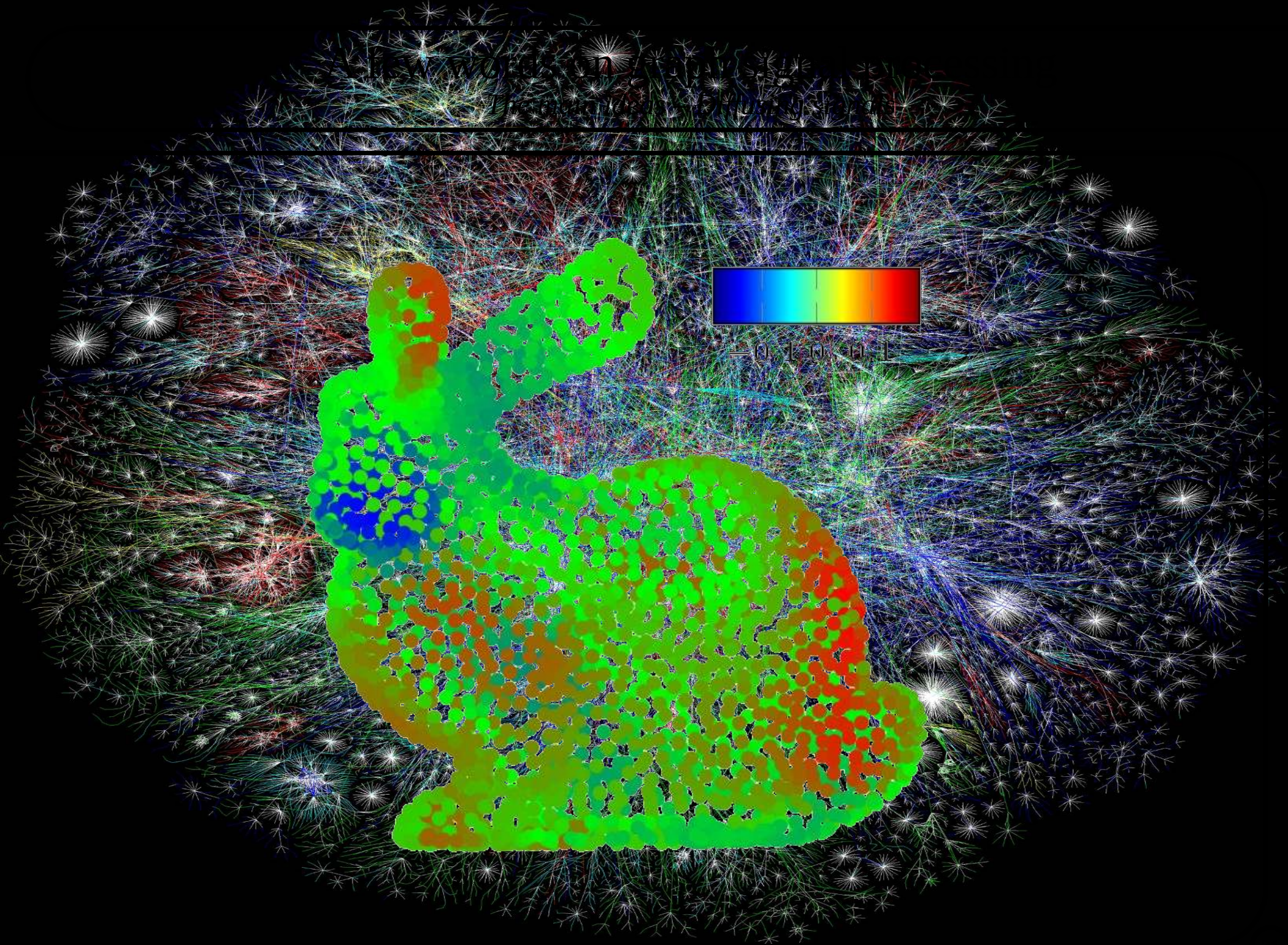


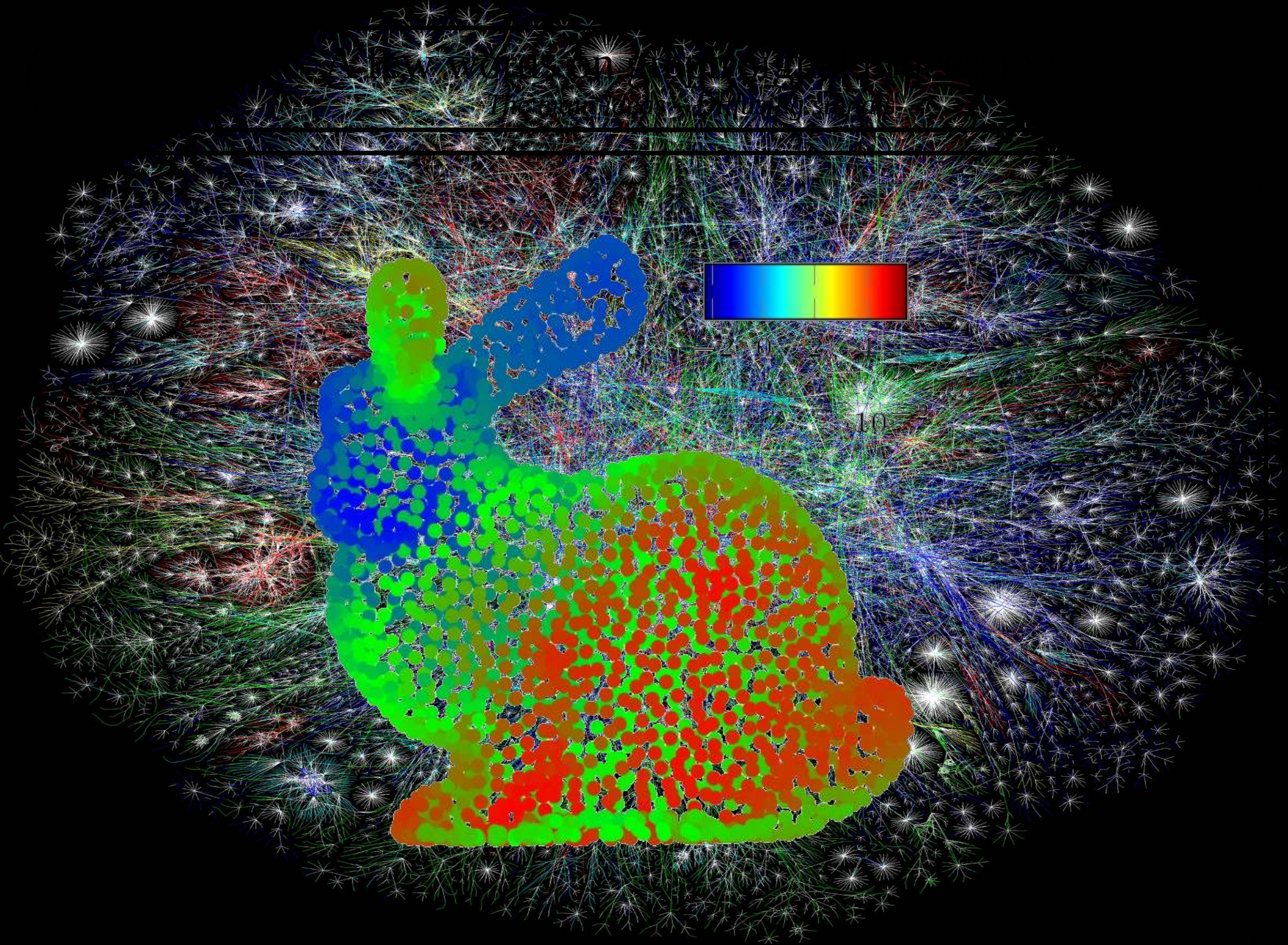
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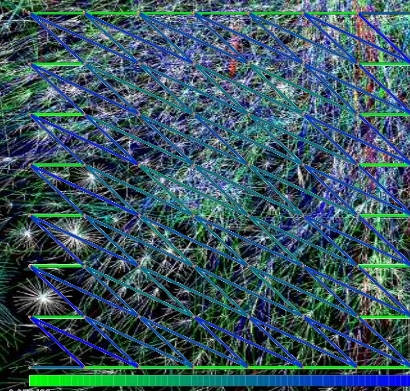
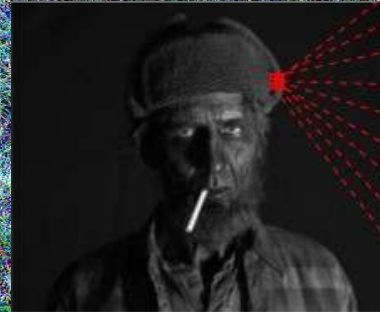




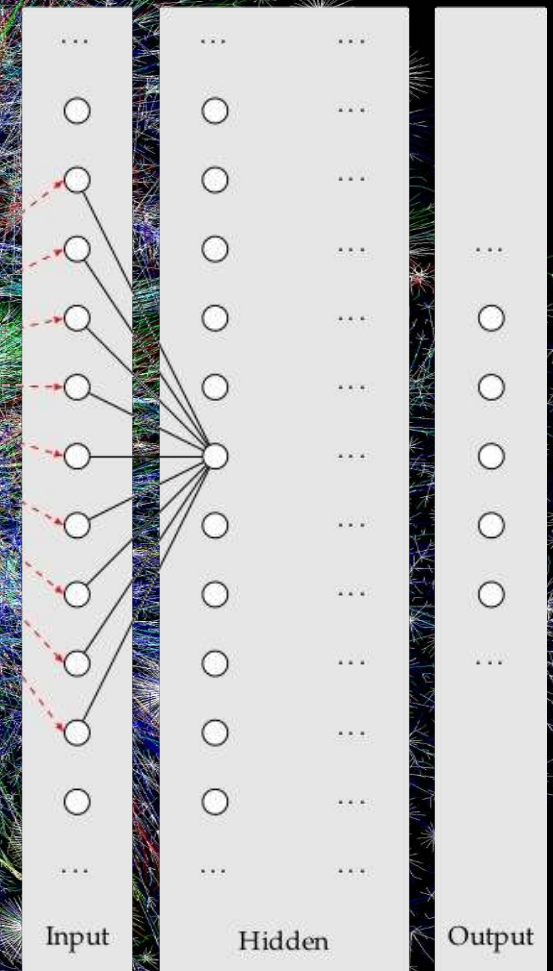
A NEW WORLD IN DATA

The new world of problems and solutions

- ▶ Denoising
- ▶ Downsampling
- ▶ Compression of signals
- ▶ Patterns recognition
- ▶ Machine learning
- ▶ Understanding the brain



0.953480



► A few words on graph signal processing

- * What is it?
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- * Some problems and applications

► Graph inference from signals

- * Why it may be necessary
- * The case of stationary signals

► Translations on irregular graphs

- * How we define them
- * How we identify them
- * Examples and applications

Graphs in the Wild

Why? How? What? When? Where? How often?

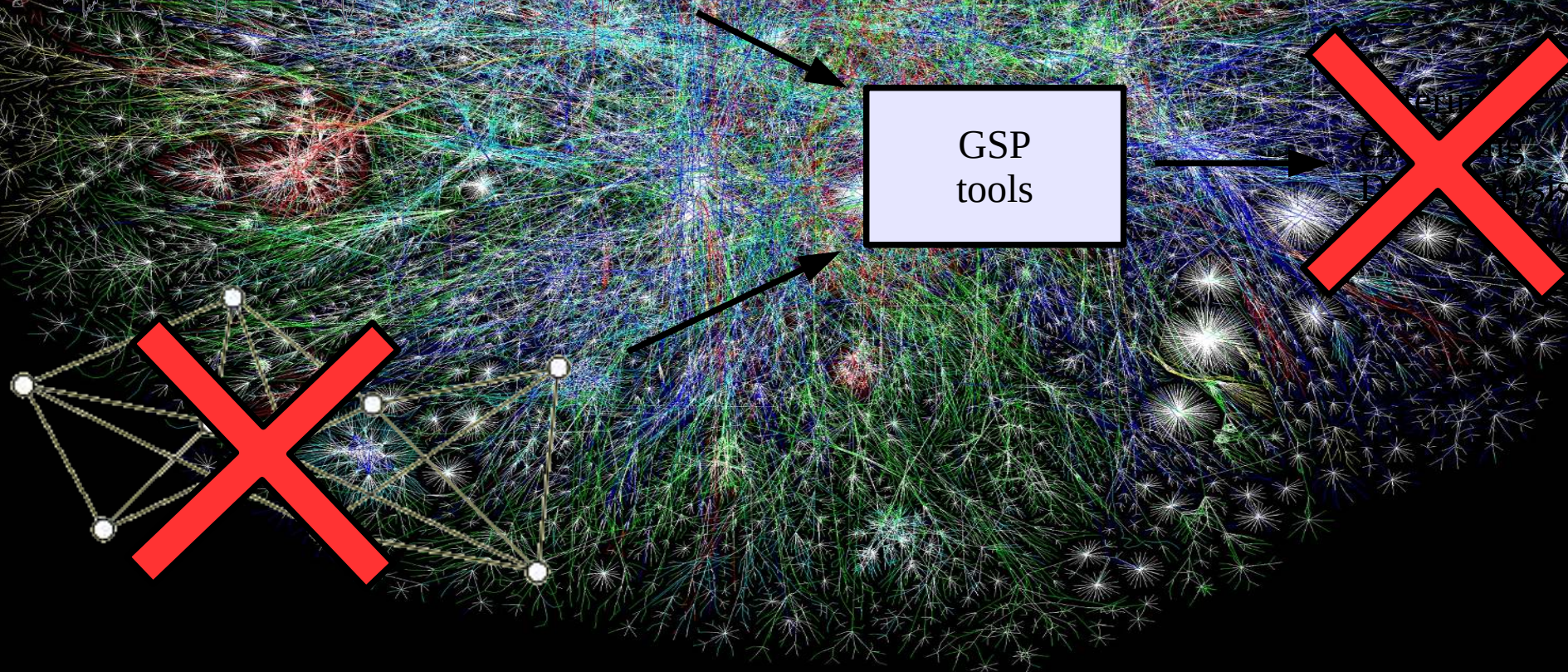
GSP
tools

Filtering
Clustering
Data analysis



Graphs in the Wild

Why? How? What? When? Where? How often?



Graphs in ML

Why it may be needed in ML

Graph
inference

GSP
tools

Filtering
Clustering
Data analysis



Graph Inference

Why it may be an essential building block

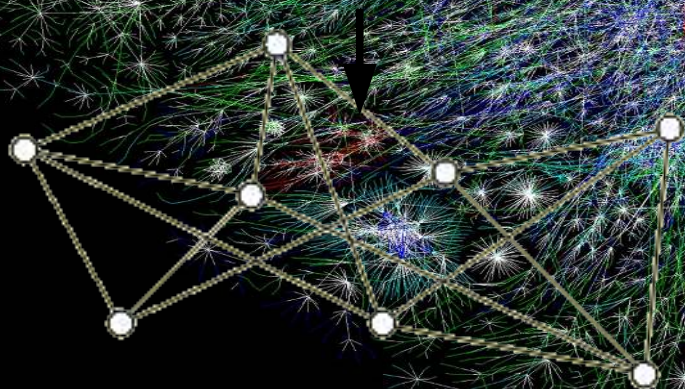
Signals:

Need for a model

Graph
inference

Graphs:

Need for assumptions



Graph Inference

Why it may be an essential building block

Signals:

Need for a model

► Stationary processes

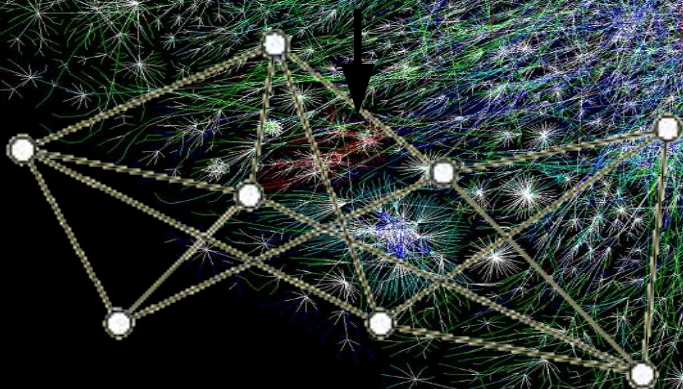
Graph
inference

Graphs:

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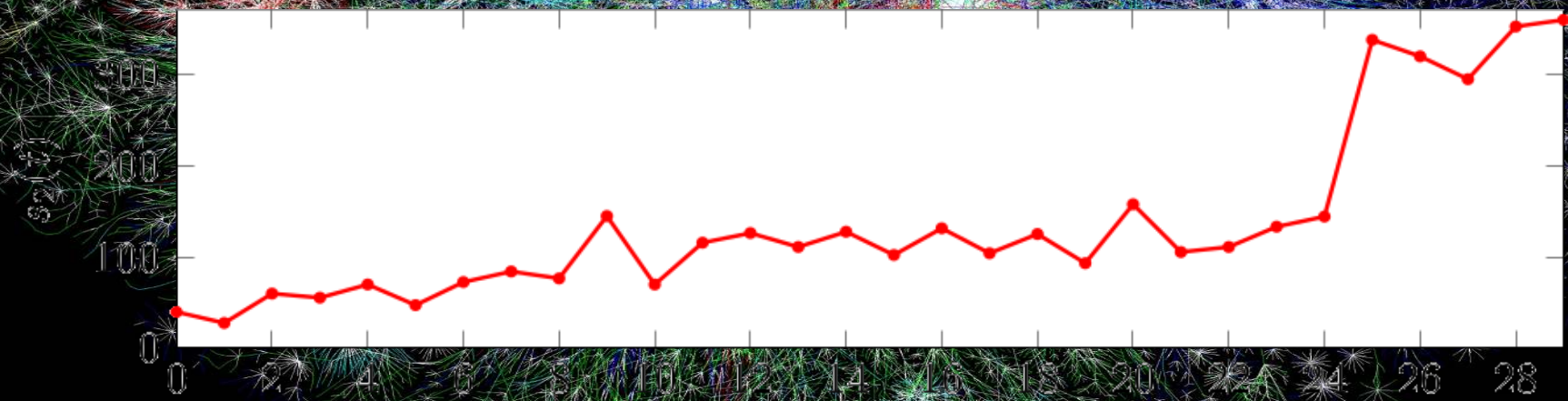
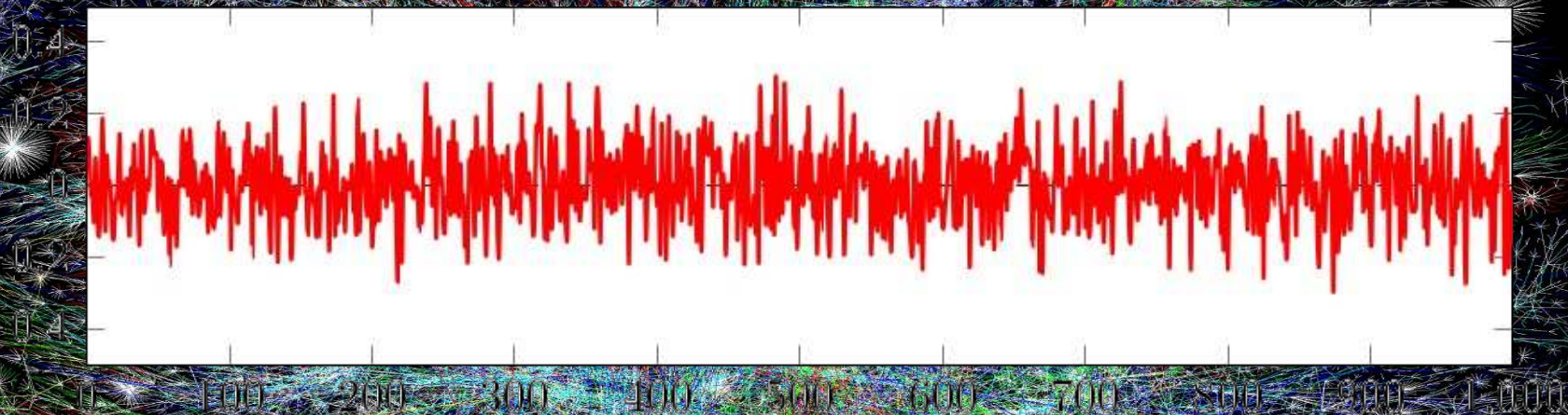
► Valid for stationary processes

► Desired features



Carapen River

Presence of stationary signals - stationary noise



Graph Filter with

The use of stationary signal conditions in graph filter

Definition

A stochastic graph signal $\mathbf{x}$ defined on the vertices of a graph G is called *graph wide (or second order) stationary* (GWSS), if and only if it satisfies the following properties:

Its first moment is constant over the vertex set: $m_{\mathbf{x}}[i] = \mathbb{E}\{\mathbf{x}[i]\} = c \in \mathbb{R}$

Its covariance is invariant with respect to the localization operator:

$$\gamma_{\mathbf{x}}[i, n] = \mathbb{E}\{(\mathbf{x}[i] - m_{\mathbf{x}})(\mathbf{x}[n] - m_{\mathbf{x}})\} = \mathcal{T}_i \gamma_{\mathbf{x}}[n]$$

Example

White noise is GWSS

If a graph filter g is applied to a GWSS signal, the result remains GWSS

Graphs and Neural Networks

The of stochastic Hopfield's solution for a set of

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph Shift Operator

Use of stationary hypothesis: diffusion on a single graph

Number of signals

Observed signals

i.i.d signals

$$\forall i \in \{1, \dots, M\} : \mathbf{x}_i = \mathbf{T}^k \mathbf{y}_i$$

Graph shift operator used
for diffusion

Variable number of diffusions
(linear combination also ok)

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Graph Shift Operators

The case of stationary signals

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Stationary signals enforces coherence between the signal model and the

graph structure

Graph filter networks

The case of stationary signals – the global shift operator

Characterization step enforces coherence between the signal model and the topology

Stationary
signals $\mathbf{X}$

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Inference

$\tilde{\mathbf{T}}$

- $\tilde{\mathbf{T}} \triangleq \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} \mathcal{X}^T$ is symmetric

- $\forall i, j \in \{1, \dots, N\}; j \neq i: T(i, j) \geq 0$

- Let χ_1 be the constant sign eigenvector in $\mathcal{X}$, $\lambda_1 = 1$.

- $\forall i \in \{1, \dots, N\}; \lambda_i \in [-1, 1]$

Graph filter design

The case of stationary signals – the covariance matrix

Characterization step enforces coherence between the signal model and the topology

Stationary
signals $\mathbf{X}$

Characterization

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graph shift operators

Inference

$\tilde{\mathbf{T}}$

Eigenvectors of the
covariance matrix

- $\tilde{\mathbf{T}} \triangleq \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^T$ is symmetric
 - $\forall i, j \in \{1, \dots, N\}: \lambda_i \geq 0, \mathbf{T}(i, j) \geq 0$
 - Let χ_1 be the constant sign eigenvector in $\mathcal{X}$: $\lambda_1 = 1$
 - $\forall i \in \{1, \dots, N\}: \lambda_i \in [-1, 1]$
- Diffusion matrix properties
- Scale factor

Graph filter networks

The case of stationary signals – for data sets of size

Characterization step enforces coherence between the signal model and the topology

Stationary
signals $\mathbf{X}$

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Space of valid
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$\tilde{\mathbf{T}}$

Eigenvectors of the
covariance matrix

Unknown

• $\tilde{\mathbf{T}} \triangleq \mathcal{X}^* \mathcal{X}$ is symmetric

• $\forall i, j \in \{1, \dots, N\} : \mathcal{X}_{ij} \geq 0, \mathbf{T}(i, j) \geq 0$

• Let χ_1 be the constant sign eigenvector in $\mathcal{X}$, $\chi_1 = 1$

• $\forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1]$

Diffusion matrix
properties

Scale factor

Graphs

The case of stationary graphs

$$\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0$$

$$\tilde{\mathbf{T}} = \mathbf{X} \begin{pmatrix} \widetilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \widetilde{\lambda}_N \end{pmatrix} \mathbf{X}^\top$$

$$\tilde{\mathbf{T}}(i, j) = \alpha_{ij1} \widetilde{\lambda}_1 + \dots + \alpha_{ijN} \widetilde{\lambda}_N$$

Graphs

The case of stationary graphs: $\lambda_1 = 0$ in the $\mathbf{A}(\mathbf{I})$

$$\forall i, j \in \{1, \dots, N\}; j \geq i : \tilde{\mathbf{T}}(i, j) \geq 0$$

$$\tilde{\mathbf{T}} = \mathbf{X} \begin{pmatrix} \widetilde{\lambda}_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \widetilde{\lambda}_N \end{pmatrix} \mathbf{X}^\top$$

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$$\forall i, j \in \{1, \dots, N\}; j \geq i : \alpha_{ij1} \widetilde{\lambda}_1 + \dots + \alpha_{ijN} \widetilde{\lambda}_N \geq 0$$

Carsten Röhl

The case of symmetric $\mathfrak{sl}_3$ - $\mathfrak{sl}_3$

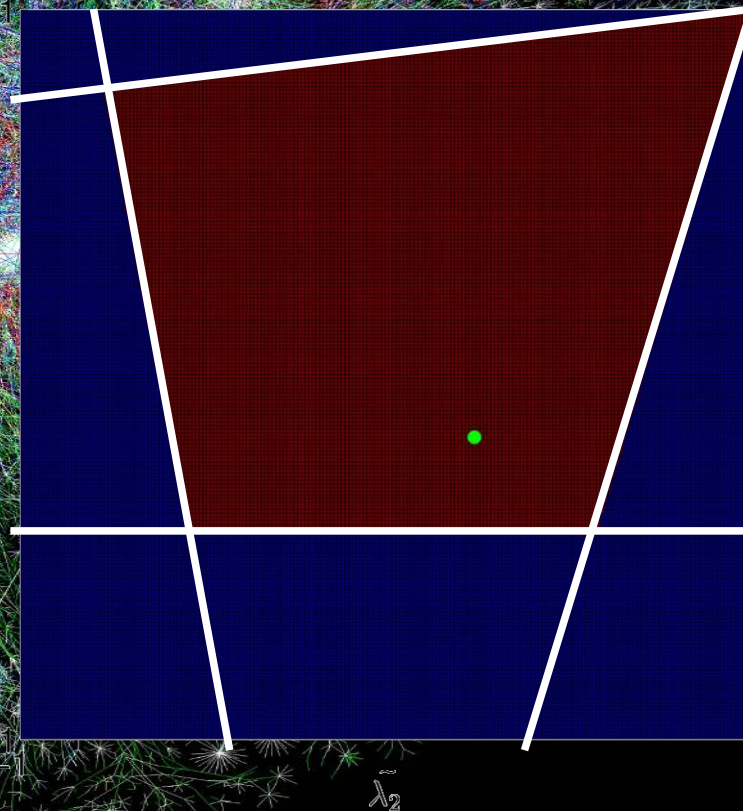
$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)}_{Q_{231}}\lambda_1 + \underbrace{\chi(2,2)\chi(3,2)}_{Q_{232}}\lambda_2 + \underbrace{\chi(2,3)\chi(3,3)}_{Q_{233}}\lambda_3$$

Graphs and Networks

The case of symmetric matrices

$$T(2,3) = \underbrace{\chi(2,1)\chi(3,1)}_{Q_{231}}\lambda_1 + \underbrace{\chi(2,2)\chi(3,2)}_{Q_{232}}\lambda_2 + \underbrace{\chi(2,3)\chi(3,3)}_{Q_{233}}\lambda_3$$

$$W = \begin{pmatrix} 0.417 & 0.302 & 0.186 \\ 0.302 & 0.147 & 0.346 \\ 0.186 & 0.346 & 0.397 \end{pmatrix}$$



Carapin, R. and V. V. Voevodin

The case of symmetric $\mathbf{T}$ and $\mathbf{W}$

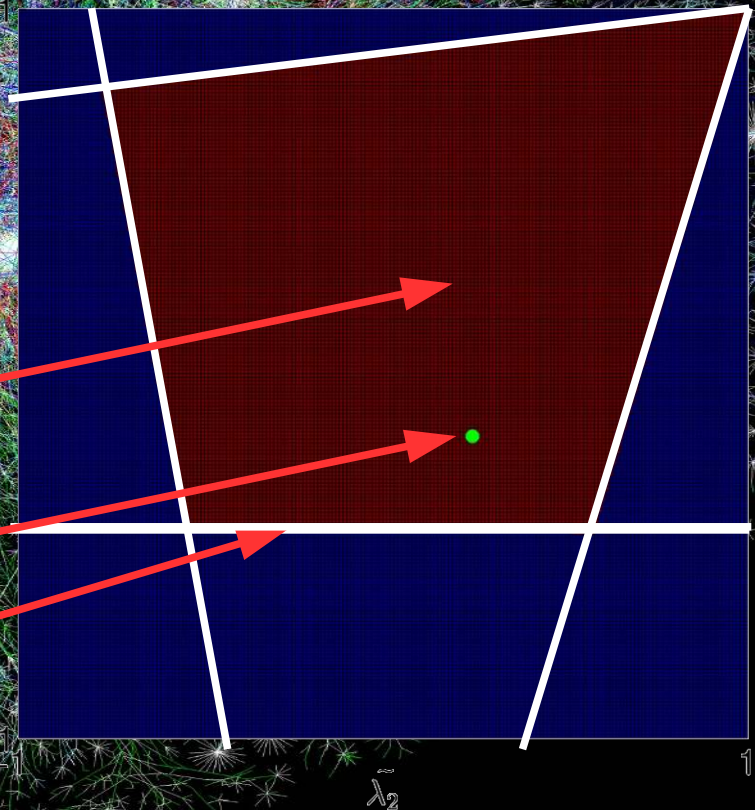
$$\mathbf{T}(2,3) = \underbrace{\chi(2,1)\chi(3,1)}_{Q_{231}}\lambda_1 + \underbrace{\chi(2,2)\chi(3,2)}_{Q_{232}}\lambda_2 + \underbrace{\chi(2,3)\chi(3,3)}_{Q_{233}}\lambda_3$$

$$\mathbf{W} = \begin{pmatrix} 0.417 & 0.302 & 0.186 \\ 0.302 & 0.147 & 0.346 \\ 0.186 & 0.346 & 0.397 \end{pmatrix}$$

Admissible $\tilde{\mathbf{T}}$

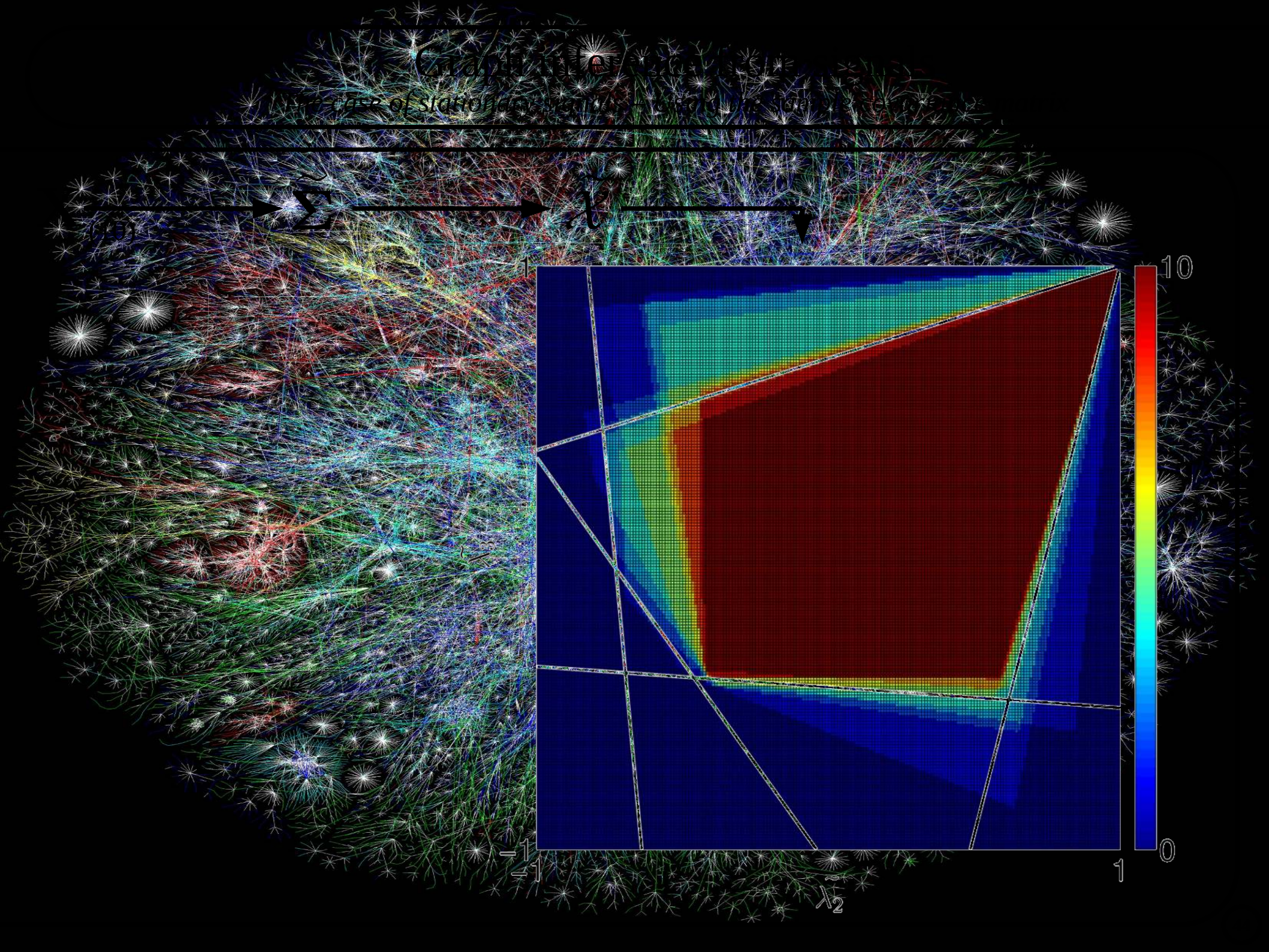
Eigenvalues of $\mathbf{T}$

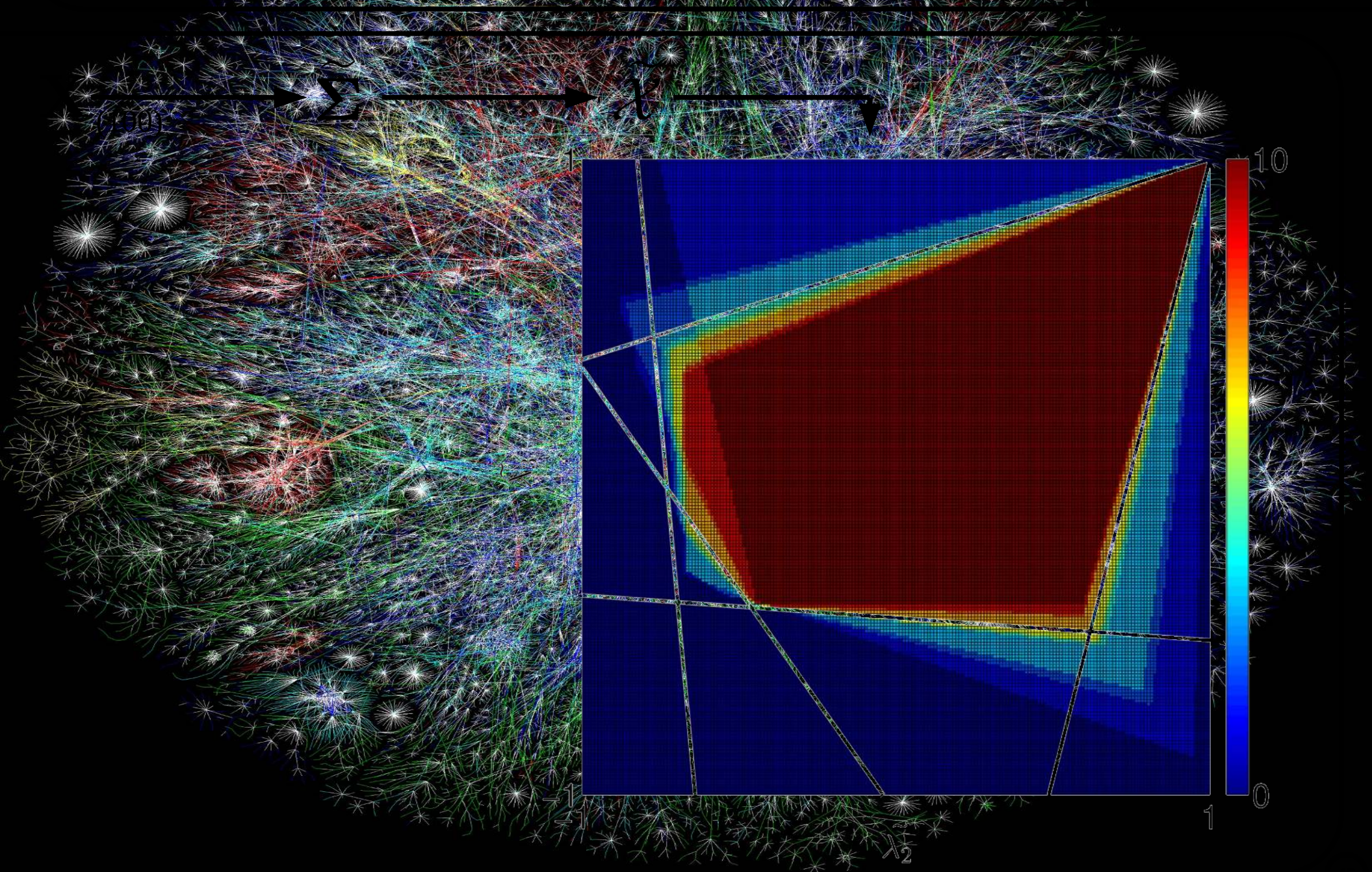
Constraints



Carapin et al.

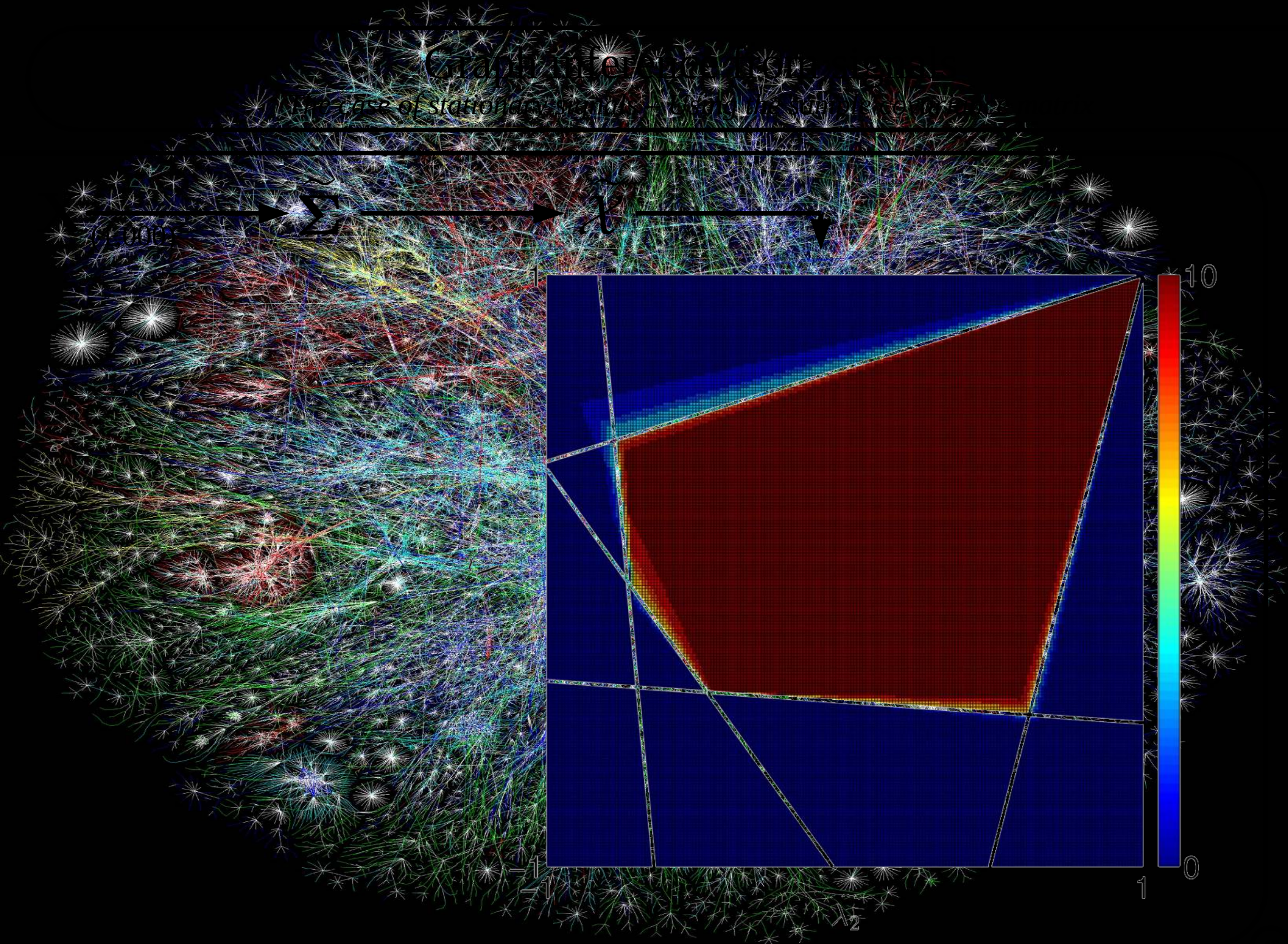
The case of stationary wavelets - time series





Carapin et al.

The case of stationary weights - case 1: the star case



Graph Inference

The case of stationary signals on networks

Inference requires selection criteria

Stationary
signals $\mathbf{X}$

Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Knowledge of the underlying topology
Knowledge of the diffusion matrix
Knowledge of the signals on the graph
Knowledge of $\mathbf{A}$ (close to) binary matrix

Graph Inference

The case of stationary signals on networks

Inference requires selection criteria

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Characterization

Space of valid
graph shift operators

Inference

$\tilde{\mathbf{T}}$

Our proposed methods

Kalotolias?

Knowledge of the underlying topology
Knowledge of the diffusion matrix
Knowledge of the signals on the graph
Knowledge of $\mathbf{A}$ (close to) binary matrix

Graphs and Networks

The case of stationary random ensembles

$$\lambda_1, \dots, \lambda_N = \arg \min_{\lambda_1, \dots, \lambda_N} \sum_{i=1}^N \lambda_i$$

$$\text{s. t. } \begin{cases} \text{Polytope constraints} \\ \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graphs and Networks

The case of stationary graphs and networks

Trace of the matrix

$$\lambda_1, \dots, \lambda_N = \arg \min_{\lambda_1, \dots, \lambda_N}$$

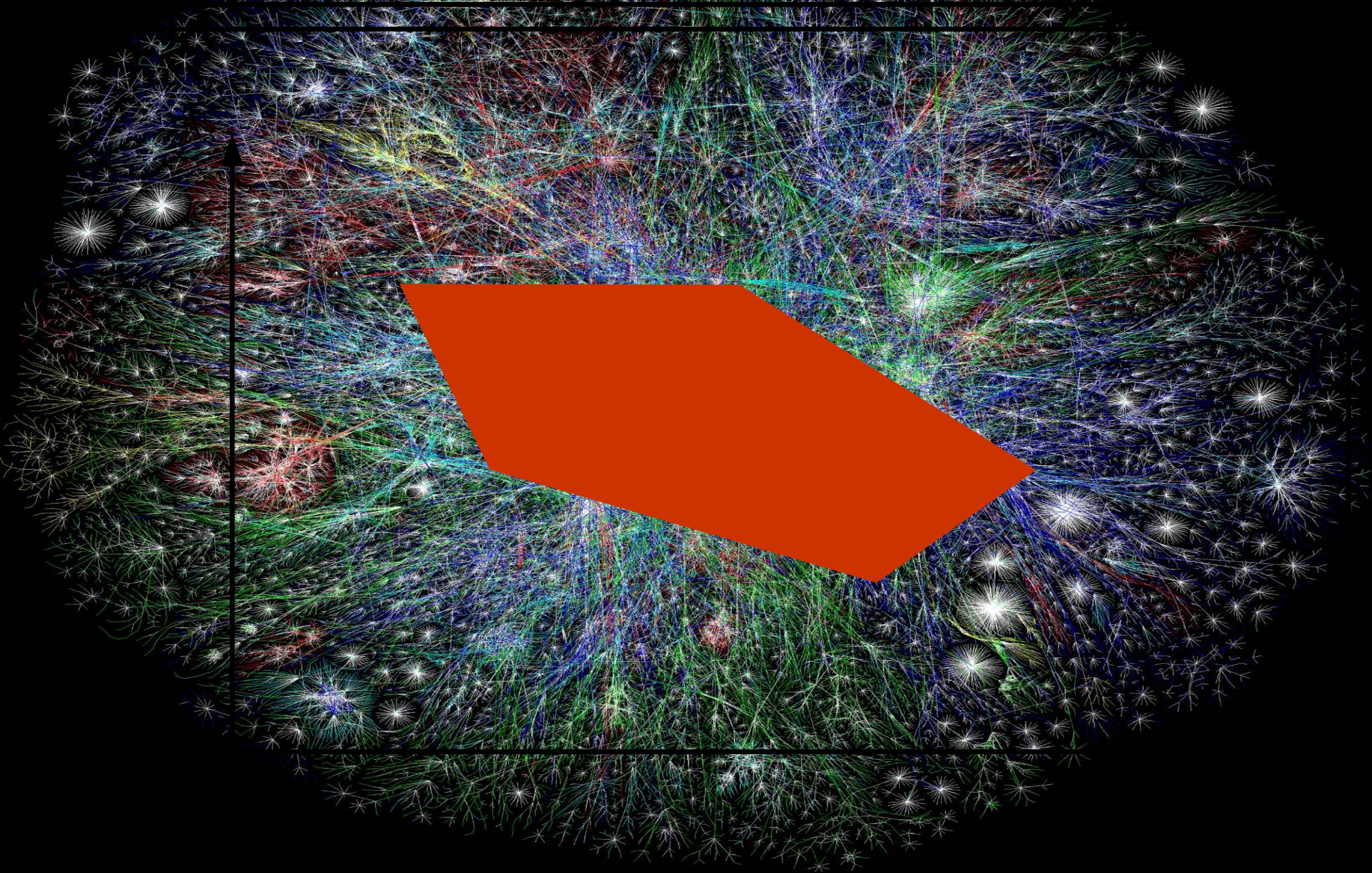
$$\sum_{i=1}^N \lambda_i$$

Polytope constraints

$$\text{s.t.} \begin{cases} \forall i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

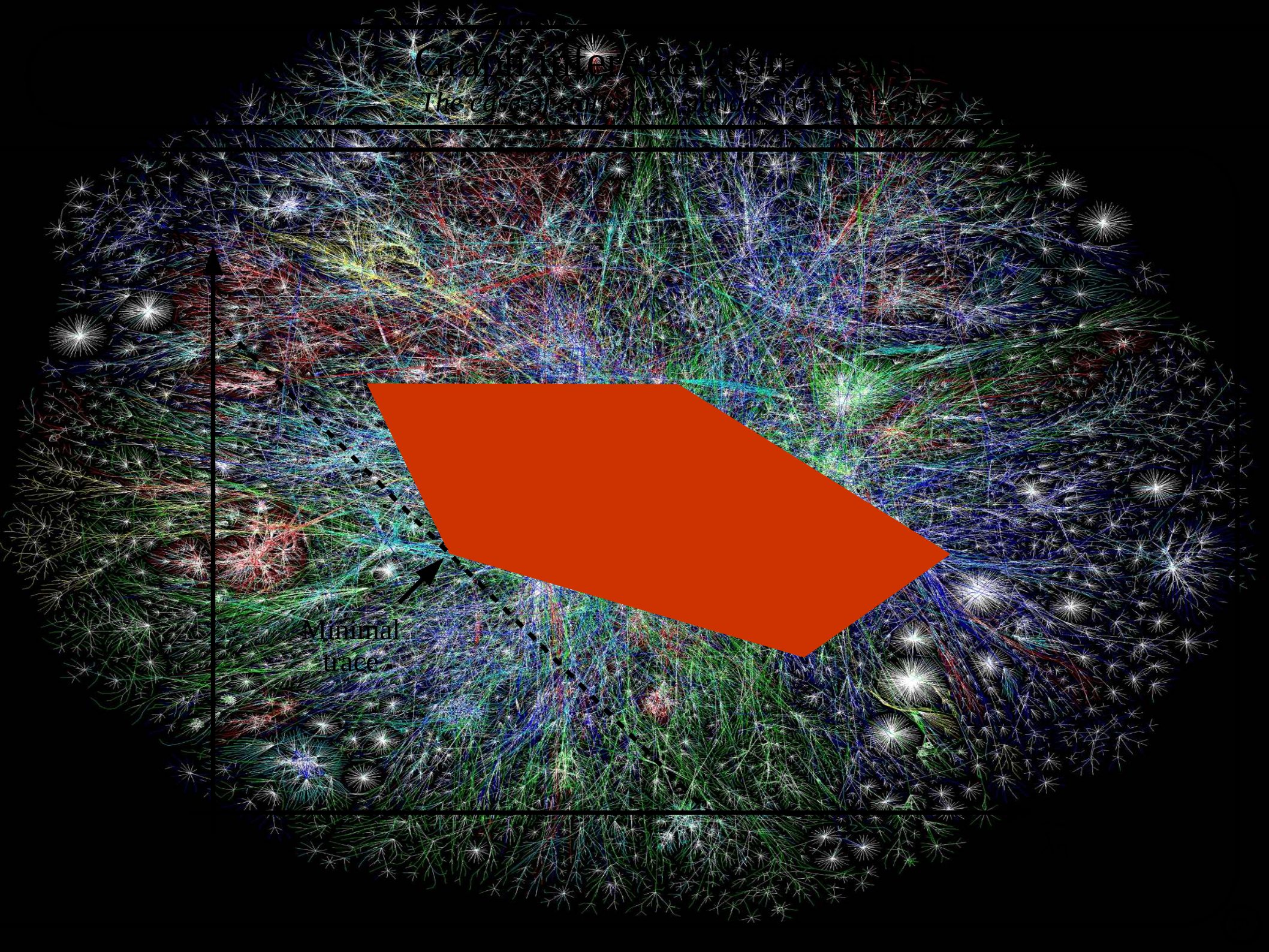
Carapen R. U. V. 2011

The case of an open system



Graphs in the Wild

The case of an open system



Graphs and Networks

The case of stationary graphs (G. L. et al., 2017)

$$[\lambda_1, \dots, \lambda_N] = \arg \min_{\lambda_1, \dots, \lambda_N} \mathbf{1}_N^\top \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^\top \mathbf{1}_N$$

$$\text{s.t.} \begin{cases} \text{Polytope constraints} \\ v_i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_i \geq 1 \end{cases}$$

Graph Partitioning

The case of stationary nodes (e.g. [1], [2], [3], [4], [5])

ℓ_1 norm

$$[\lambda_1, \dots, \lambda_N] = \arg \min_{\lambda_1, \dots, \lambda_N}$$

$$\mathbf{1}_N^\top \mathcal{X} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \dots & 0 \\ 0 & 0 & \lambda_N \end{pmatrix} \mathcal{X}^\top \mathbf{1}_N$$

$$\text{s.t.} \begin{cases} \text{Polytope constraints} \\ v_i \in \{1, \dots, N\} : \lambda_i \in [-1, 1] \\ \lambda_1 = 1 \end{cases}$$

Graph Embedding

The case of stationary and/or random walk

$\mathbf{X}$

Inference method
(eg. Kalofolias)

$$\hat{\mathbf{T}}_K = \hat{\mathbf{X}}_K \hat{\mathbf{\Lambda}}_K \hat{\mathbf{X}}_K^T$$

Graphs in the wild

The case of stationary and/or random graphs

$\mathbf{X}$

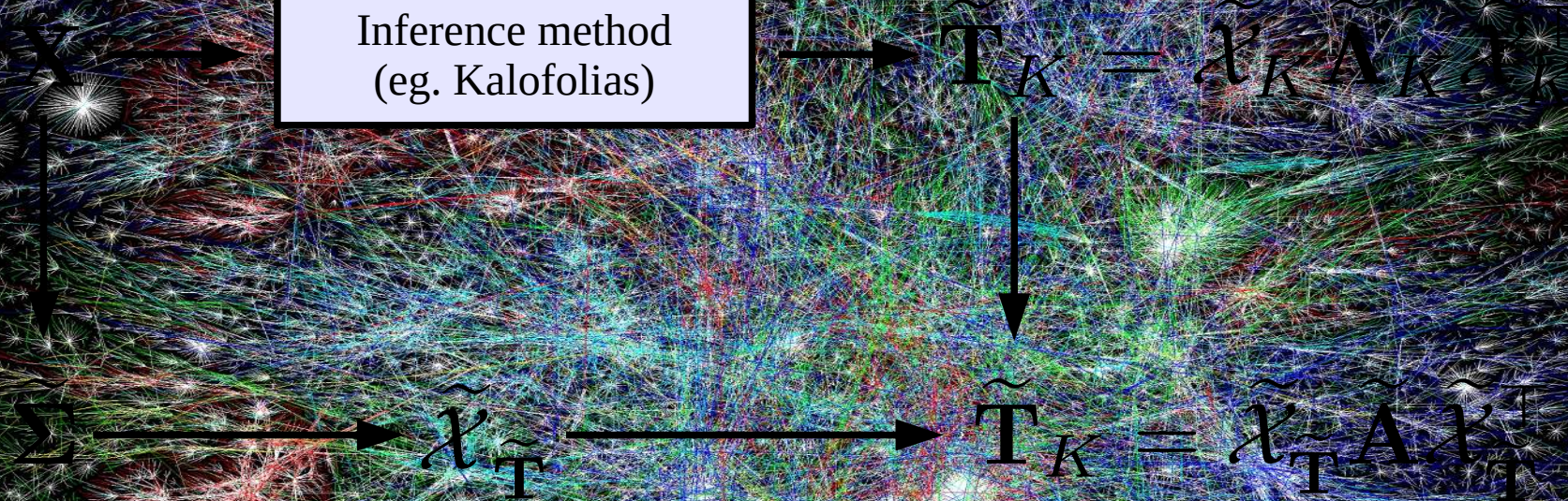
Inference method
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Not those of the
covariance matrix

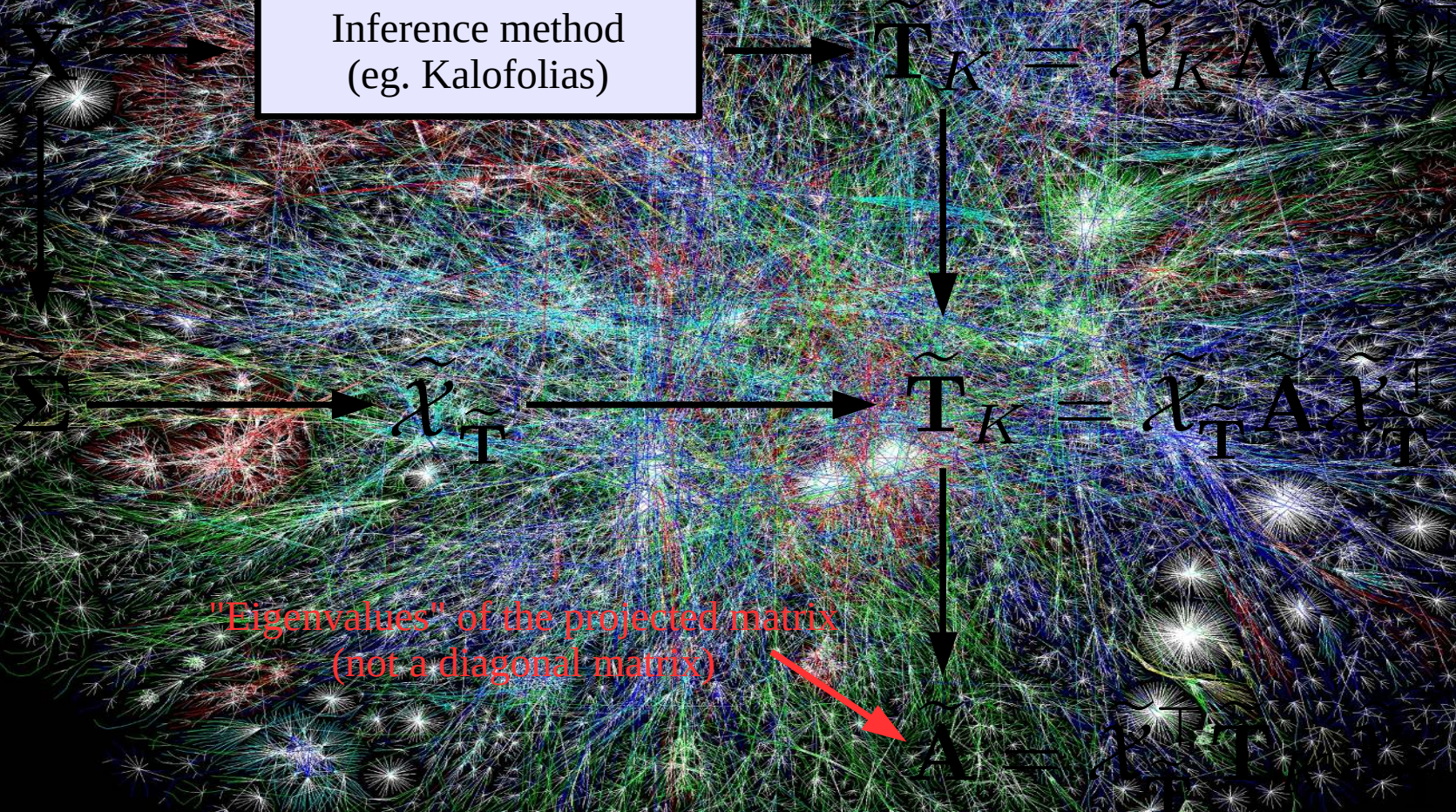
Graph Embedding

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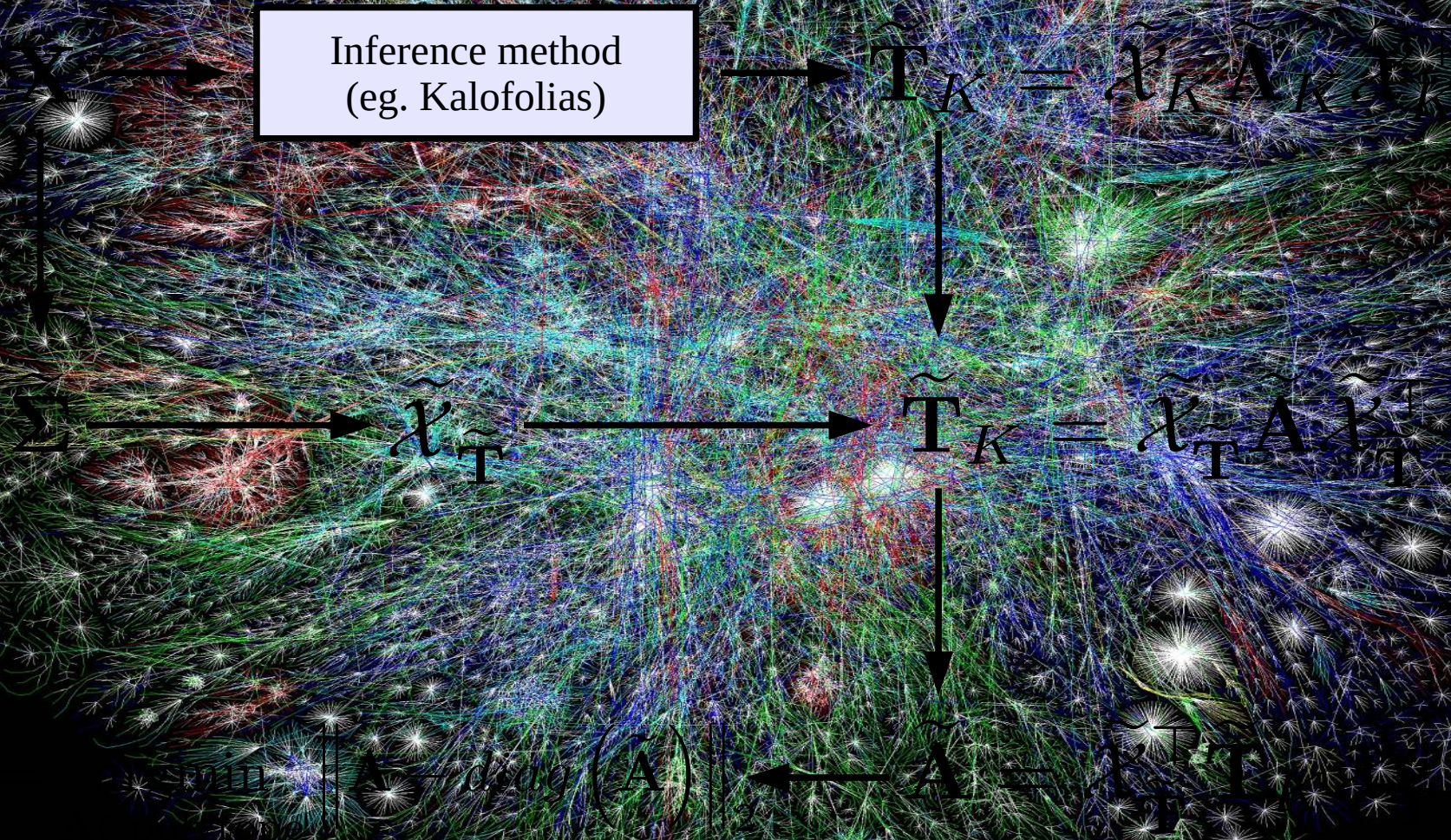
Graph Embedding

The case of stationary and/or random walks



Graph Embedding

The case of stationary and/or random walks



Graph Inference

The case of stationary and/or random graphs

Inference method
(eg. Kalofolias)

$\mathbf{X}$

$$\hat{\mathbf{T}}_K = \hat{\mathbf{X}}_K \hat{\mathbf{A}}_K \hat{\mathbf{X}}_K^T$$

Σ

$$\tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}$$

$$\tilde{\mathbf{T}}_K = \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}} \tilde{\mathbf{A}} \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}^T$$

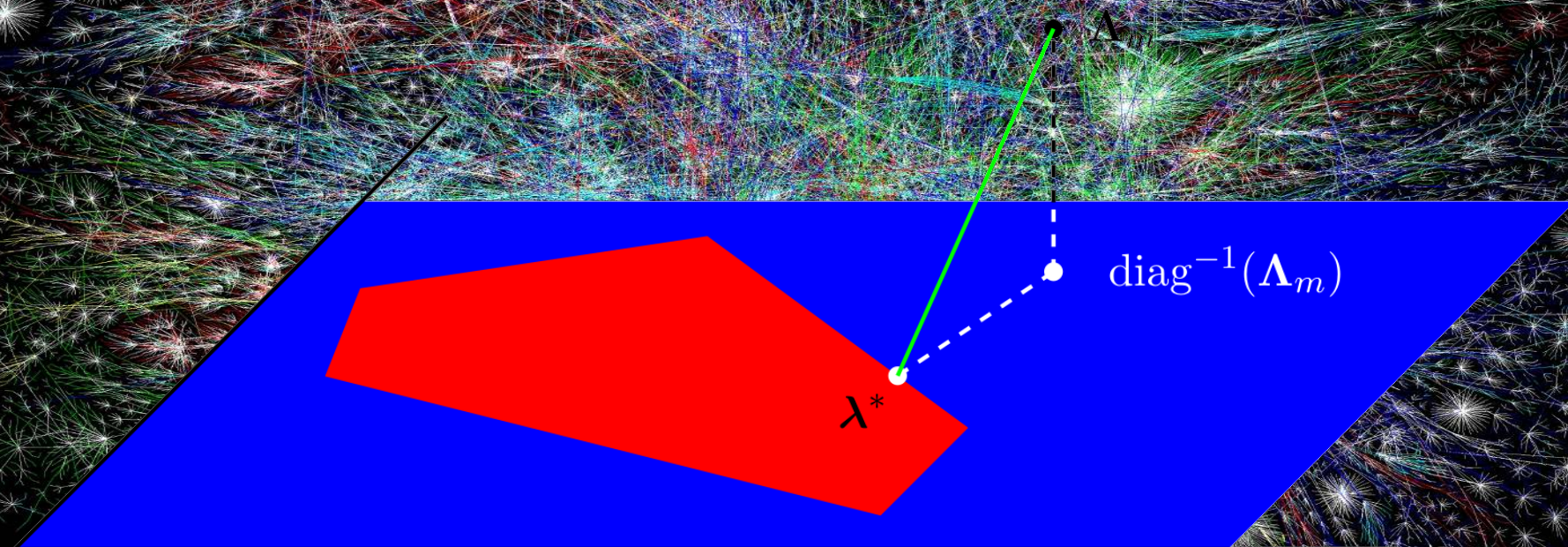
$\mathbf{T}$

$$\min_{\tilde{\mathbf{A}}} \|\mathbf{A} - \text{diag}(\tilde{\mathbf{A}})\|$$

$$\tilde{\mathbf{A}} = \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}^T \tilde{\mathbf{T}}_K \tilde{\mathbf{X}}_{\tilde{\mathbf{T}}}$$

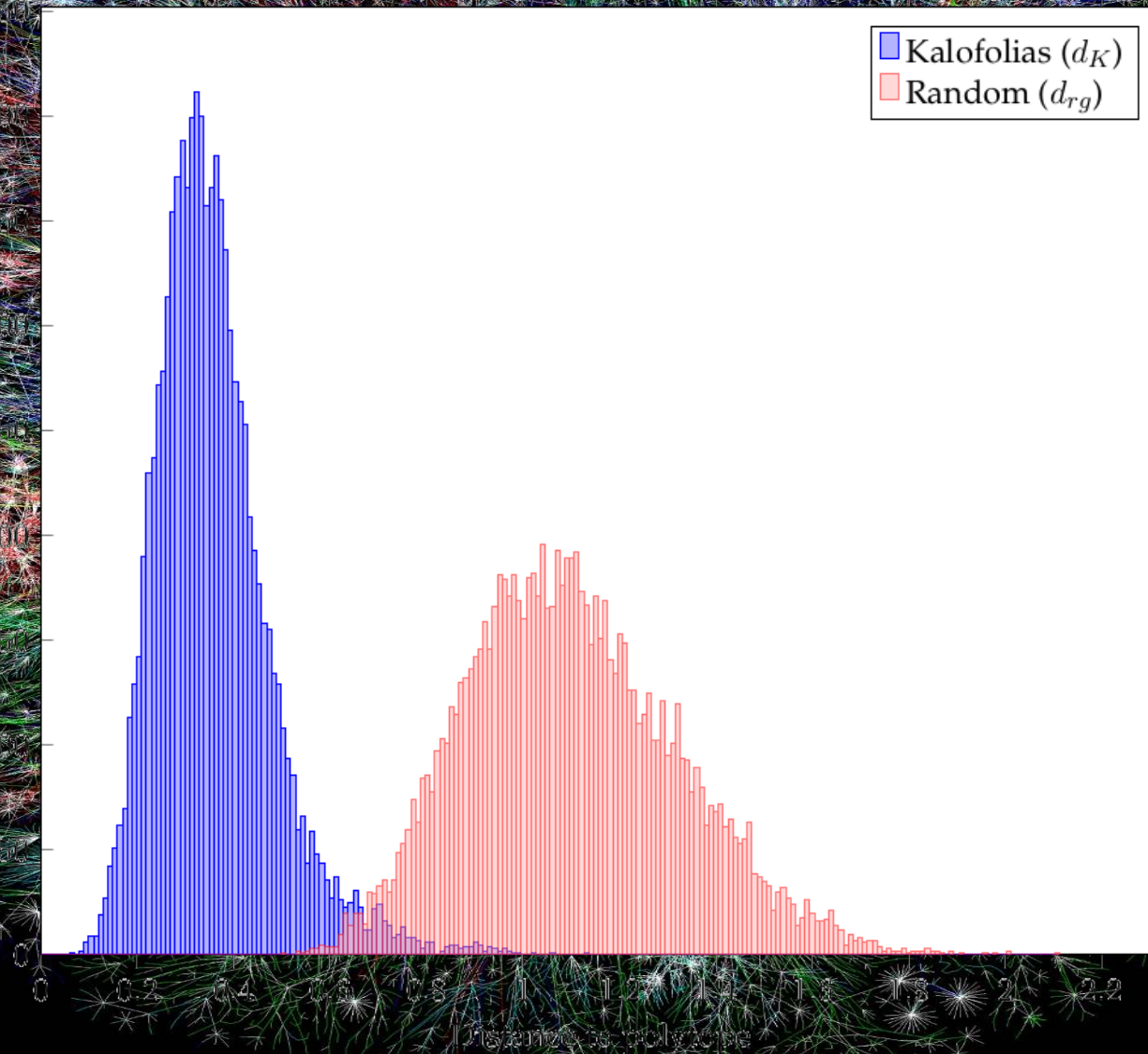
Carapin et al. (2017)

The case of an open system

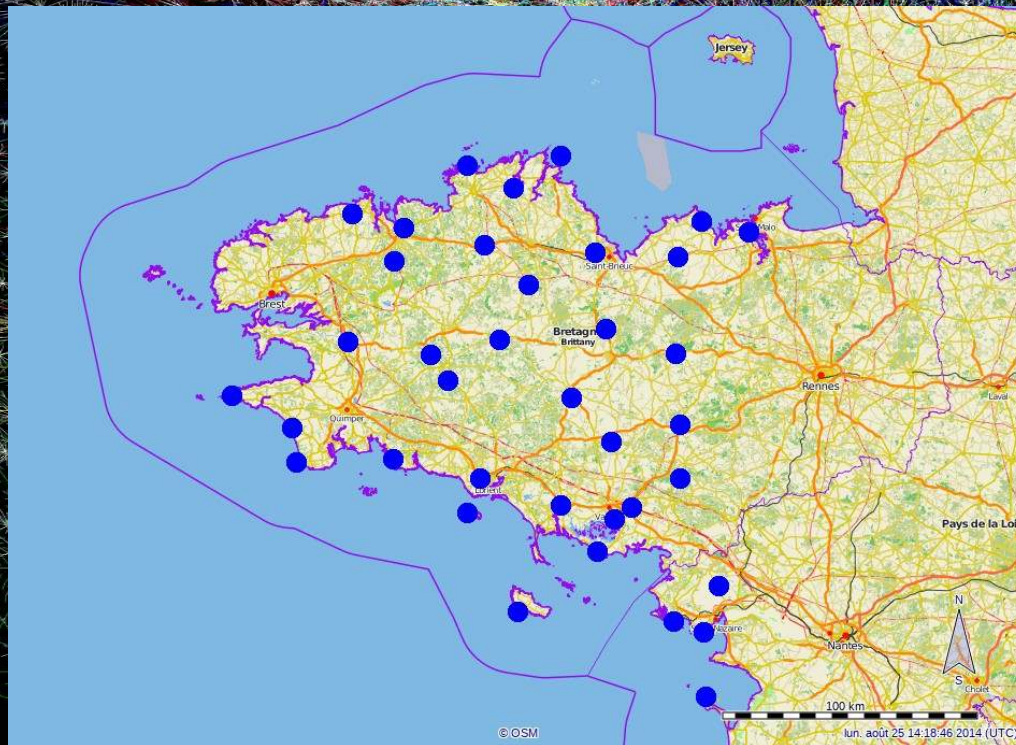


Kalofolias

The case of Salicornia



Graph of the network of the case of stations in Brittany

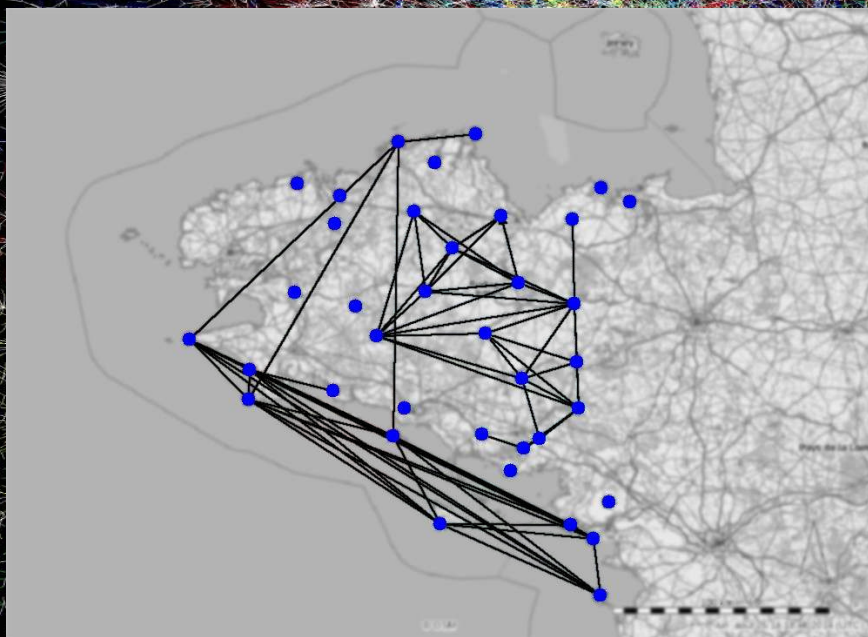


37 stations after preprocessing

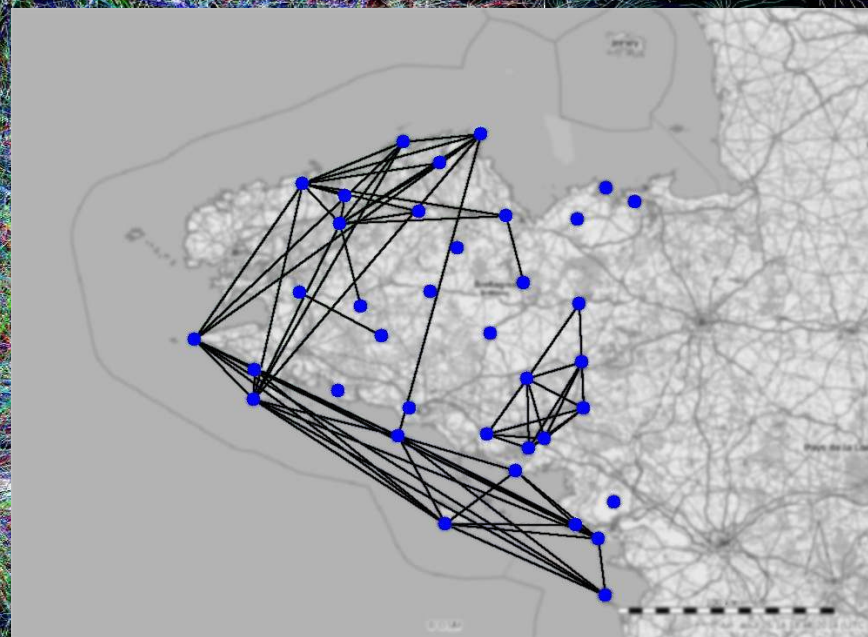
744 observations per station

Graphs

Use of stationary and non-stationary



Kalman



Regularized Kalman

Graphical Lasso

Choice of stationary variables according to $S(X)$

	polytope	$L_{1,1}$	Tr	$S(X)$
Simple	✓	36.9974	0.0013	0.0551
Sparse	✓	36.9971	0.9093	0.0585
Kalofolias	0.0313	36.9979	0	0.0751
Kalofolias closest	✓	36.9974	0.0298	0.0548
Segarra <i>et al.</i>	0.0062	36.9993	1.97×10^{-5}	0.0245
Segarra <i>et al.</i> closest	✓	36.9974	0.0046	0.0551
Graphical lasso	1.3730	35.3539	13.4977	32.8421
Graphical lasso closest	✓	36.9984	13.3584	0.0335

► A few words on graph signal processing

- * What is it?
- * The main ideas
- * Some problems and applications

► Graph inference from signals

- * Why it may be necessary
- * The case of stationary signals

► Translations on irregular graphs

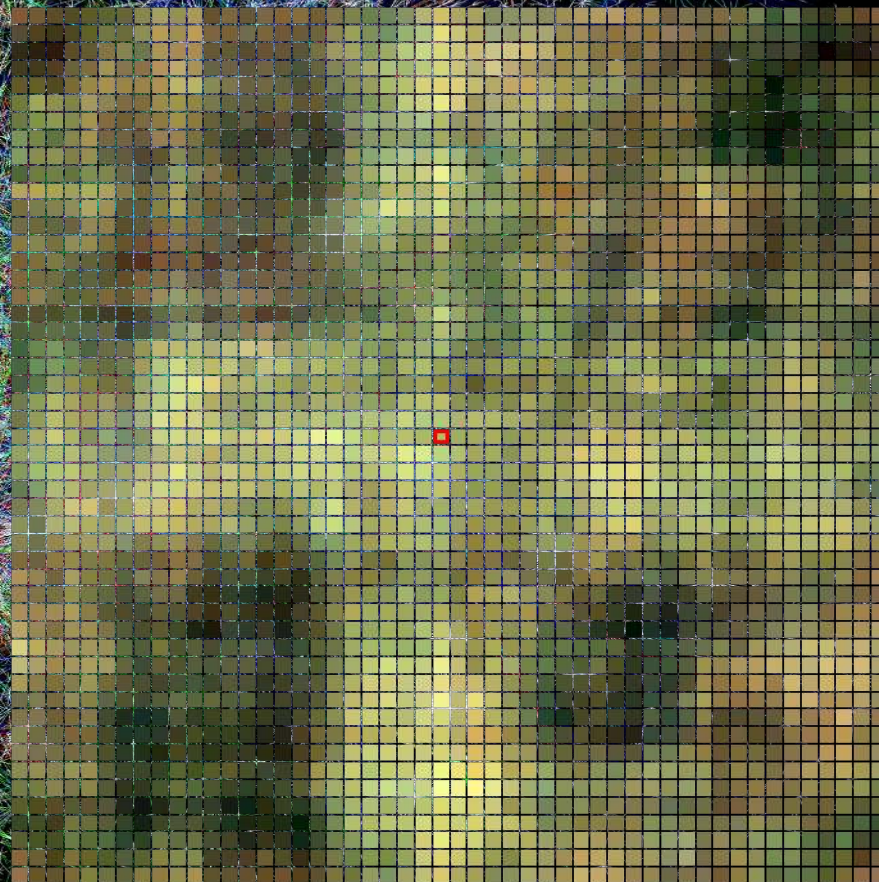
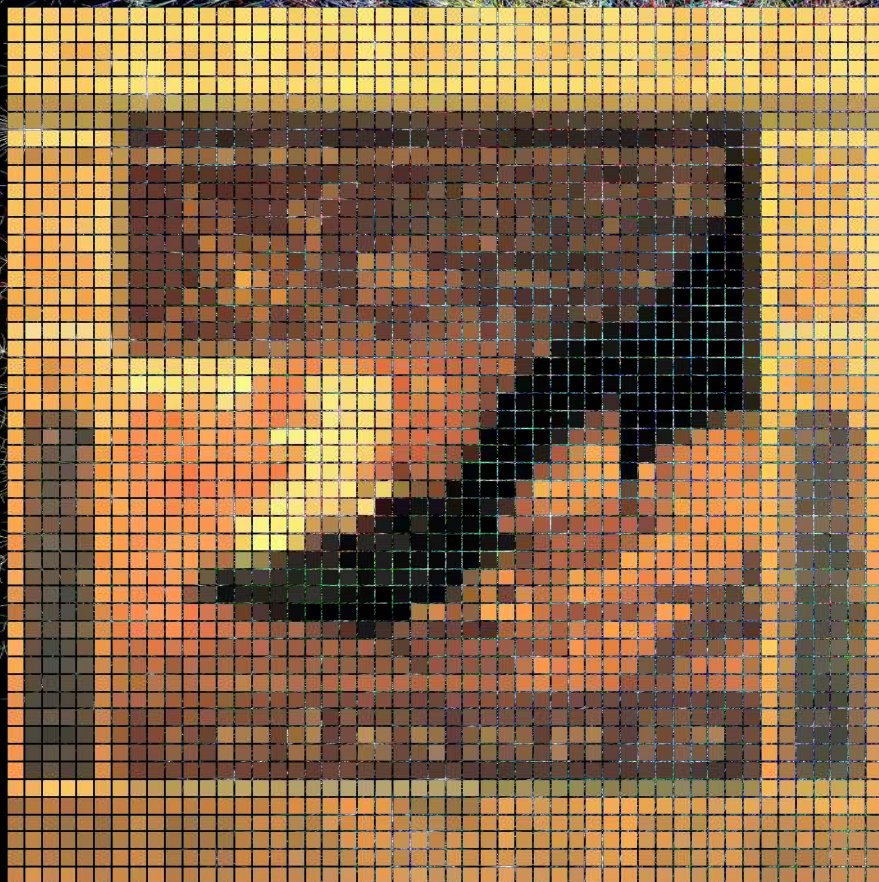
How we define them

How we identify them

How we use them and applications

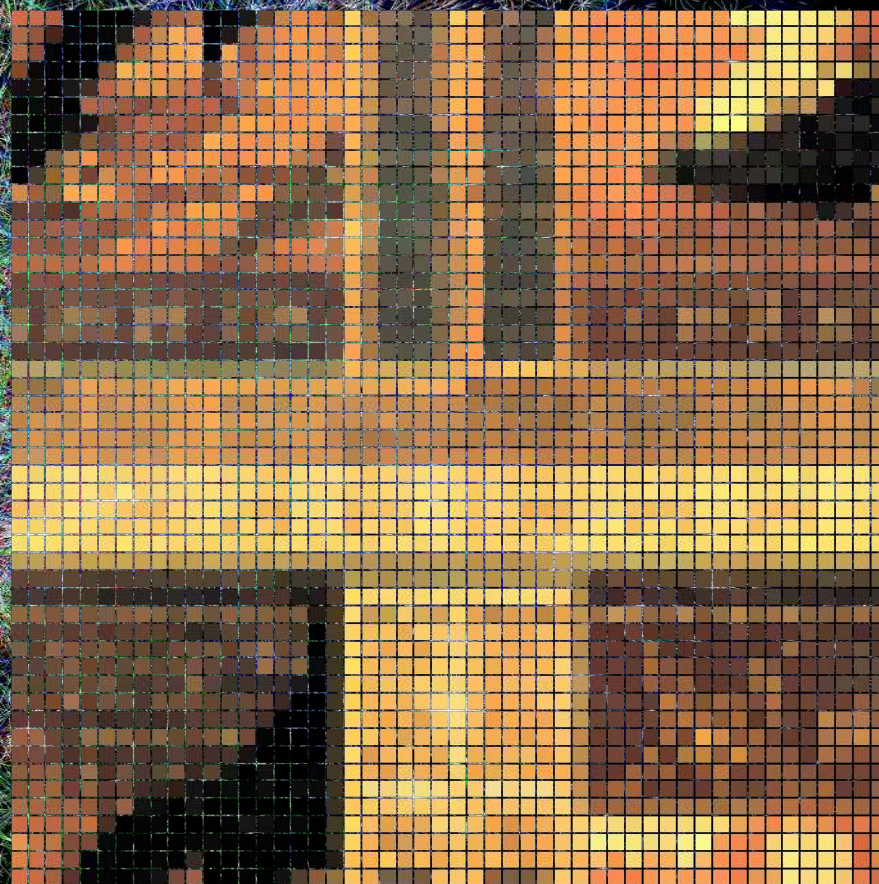
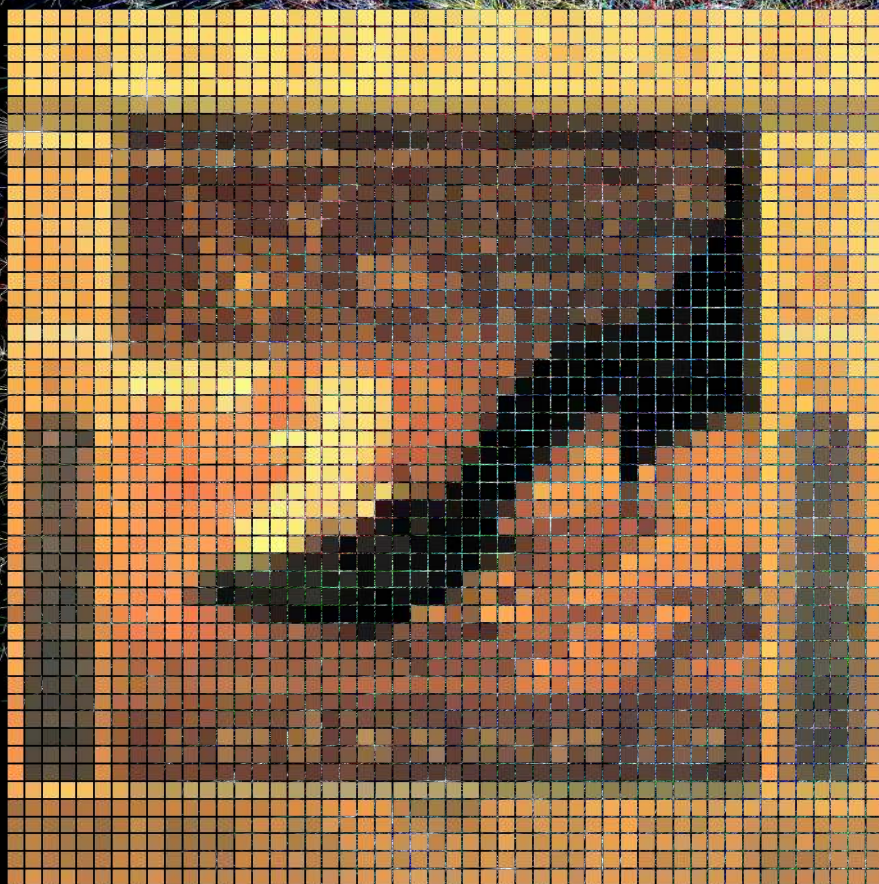
Fractal Images

How to define fractal images on a grid



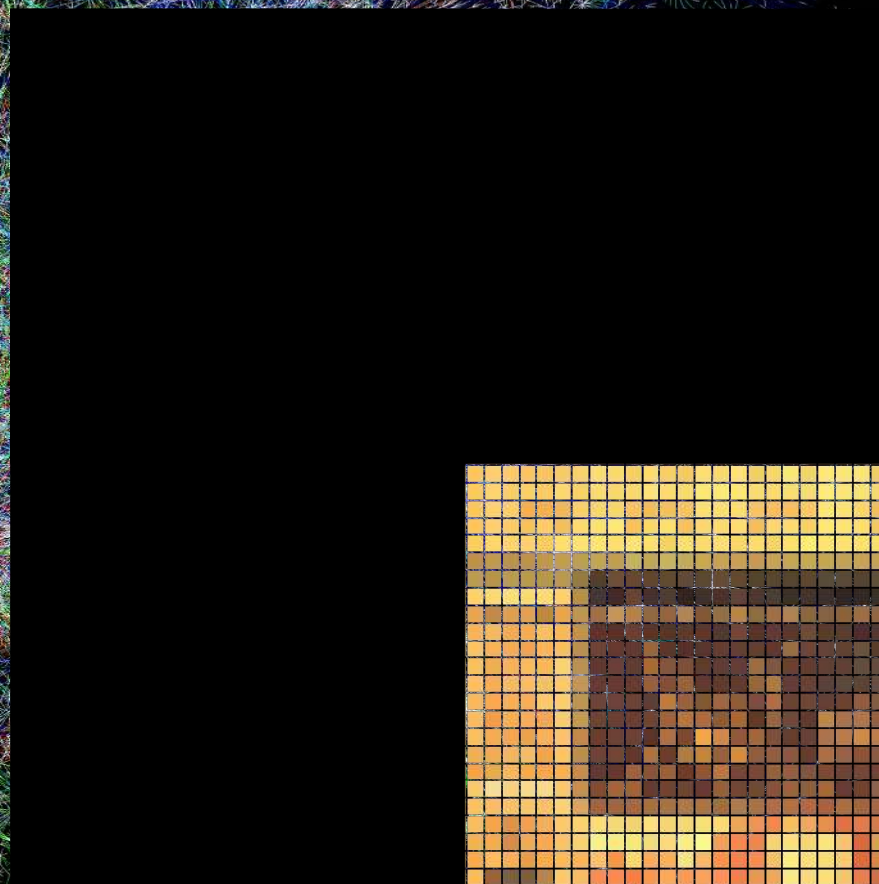
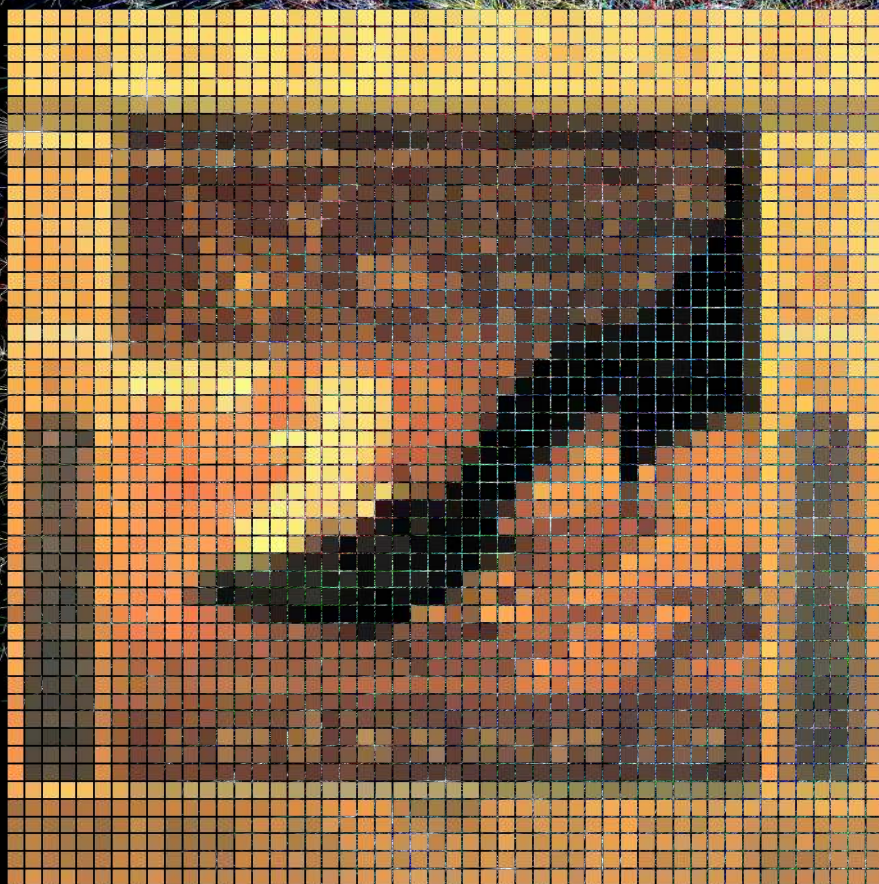
11.0115 - 11.0115

How to design a presentation



Fractal Images

How to define fractal images



Translations

How to represent a graph in a graph

Injectivity

$$\forall v_1, v_2 \in V : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

Restriction to the edges

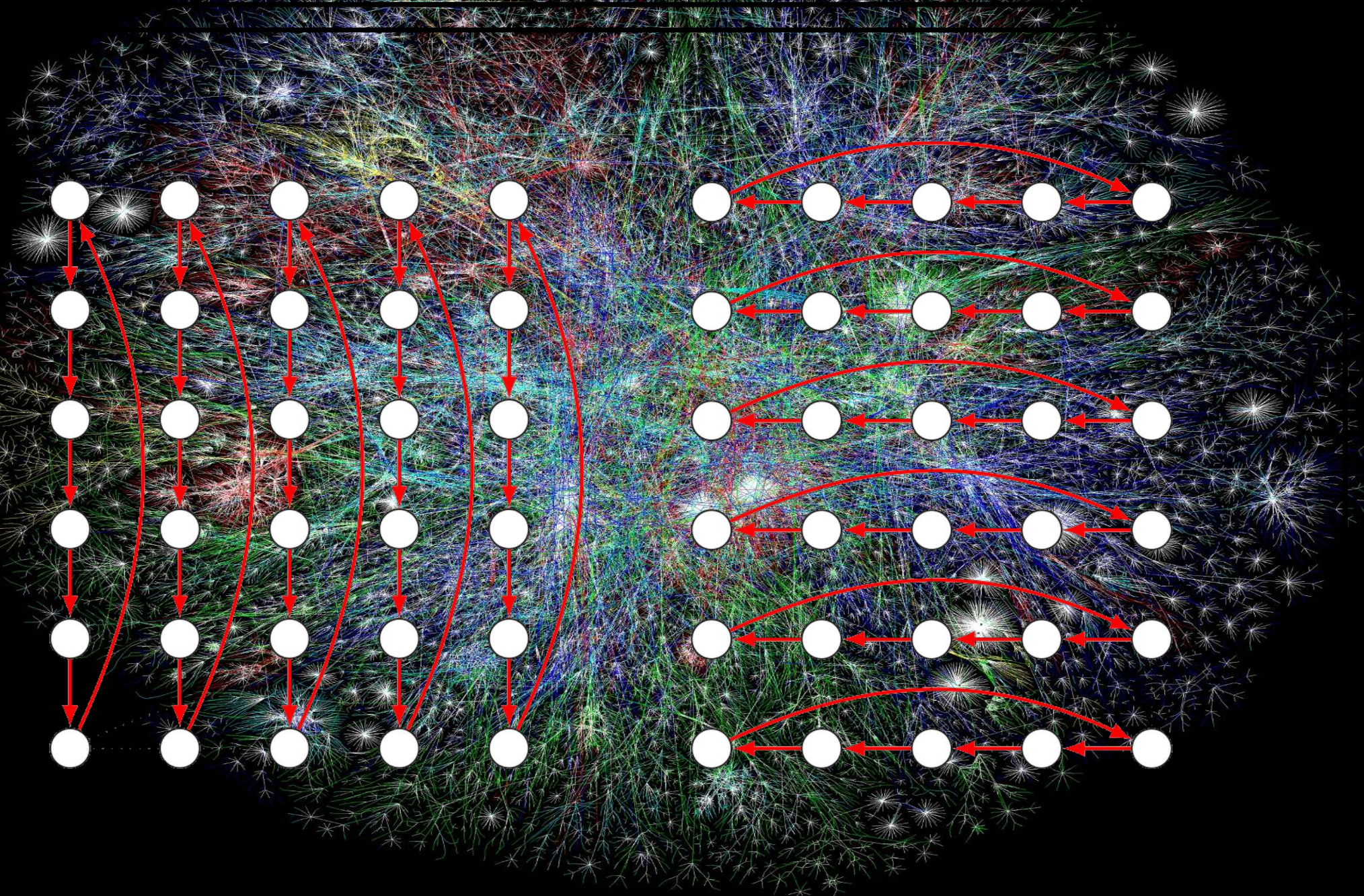
$$\forall v \in V : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

Preservation of the neighborhood

$$\forall v_1, v_2 \in V : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E} \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp))$$

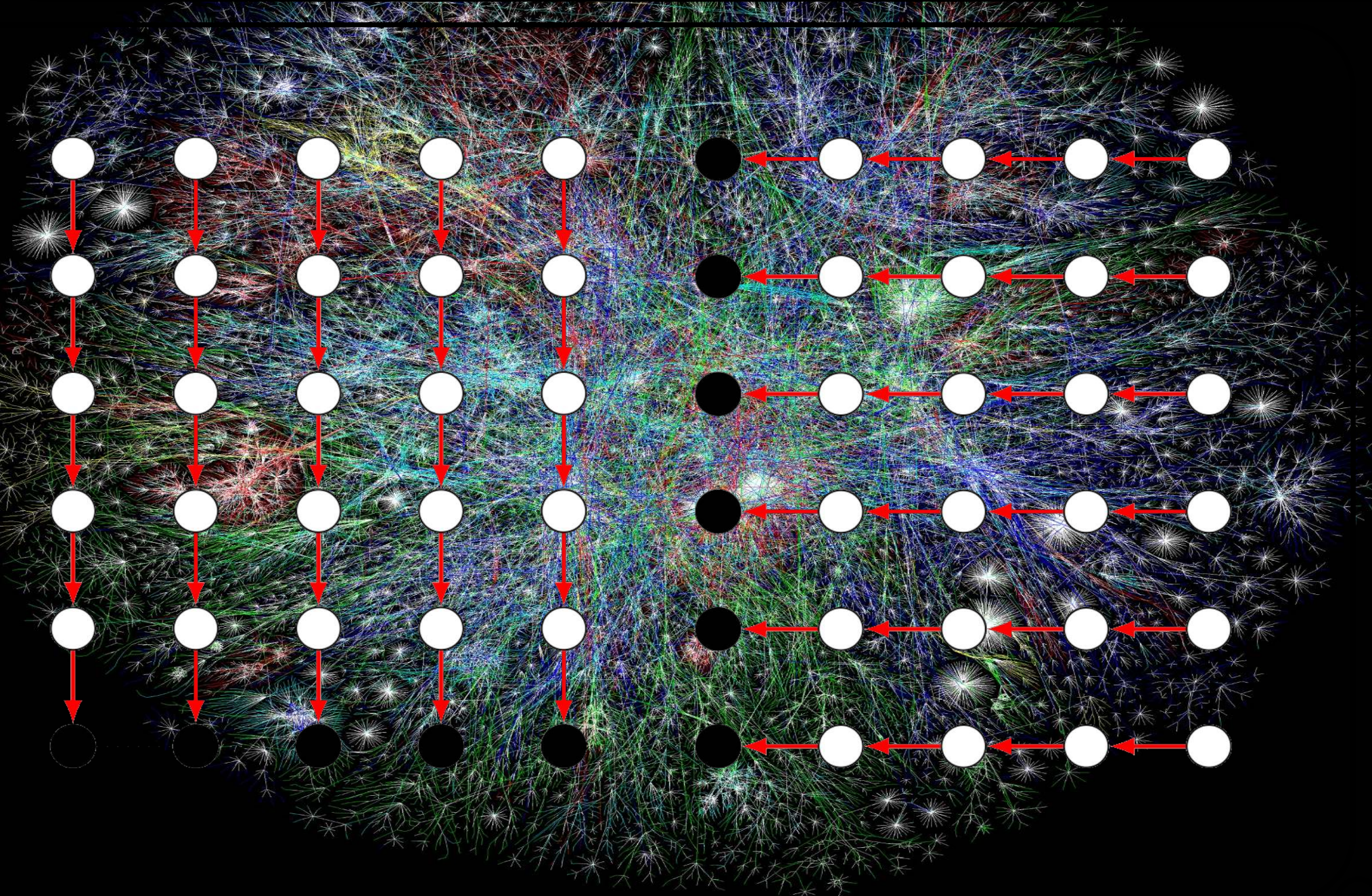
Transitions

How to identify them? How to calculate them?



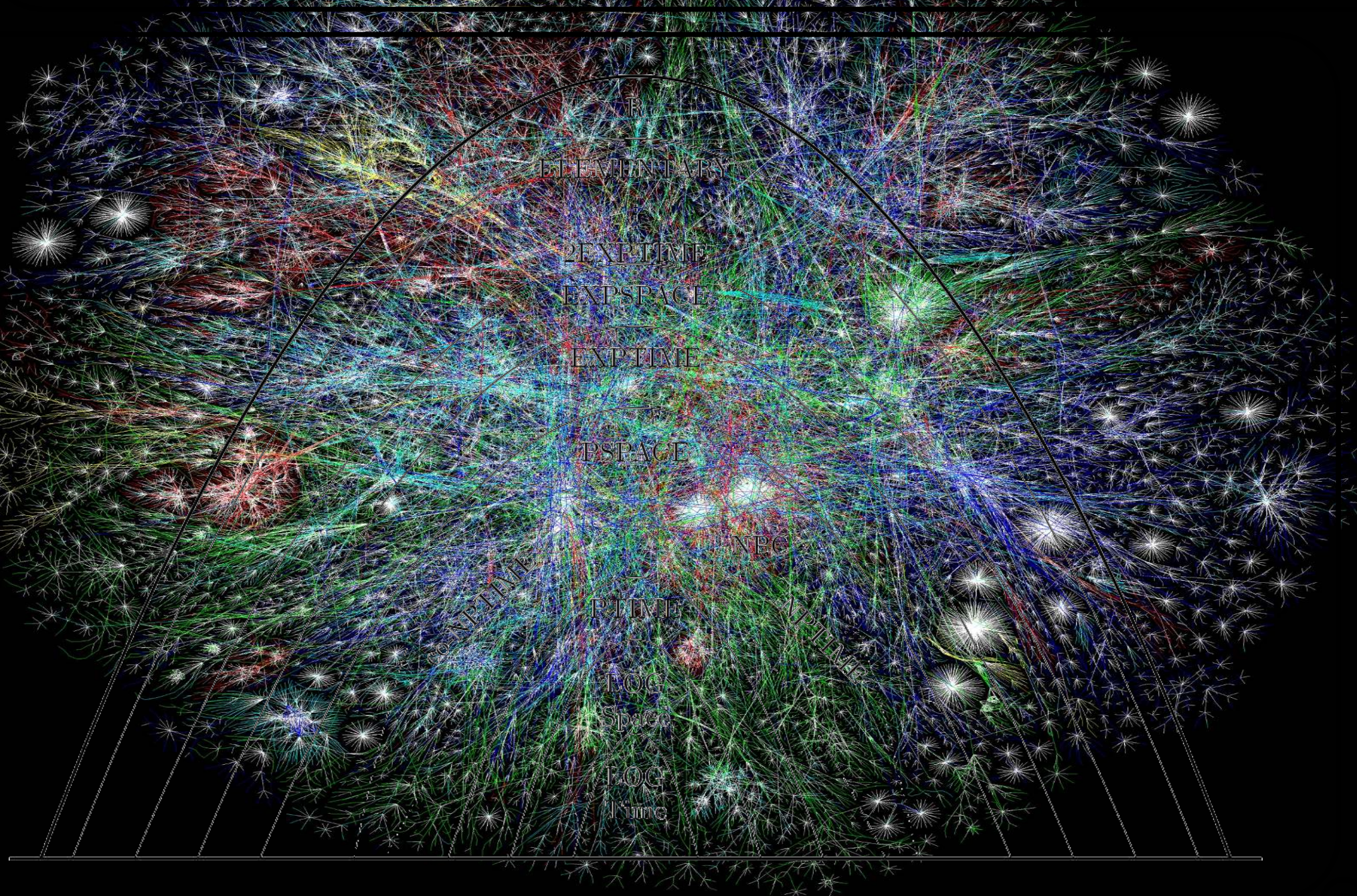
Transitions

How to identify them - I will show you a few ways



Transcendental

How to identify them - identifying the individual



Translocation

How to identify translocation events in a cell

$$\mathbf{A} \rightarrow \arg \min_{\mathbf{A}_\psi \in \{0,1\}^{N \times N}} \|\mathbf{A} \cdot \mathbf{A}_\psi - \mathbf{A}_\psi \cdot \mathbf{A}\|_1$$

Minimize deformation

$$\left\{ \begin{array}{l} \forall i, j \in [1, N] : \mathbf{A}_\psi[i, j] \leq \mathbf{A}[i, j] \\ \forall i \in [1, N] : \sum_{j=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \forall j \in [1, N] : \sum_{i=1}^N \mathbf{A}_\psi[i, j] \leq 1 \\ \|\mathbf{A}_\psi\|_1 \leq C \end{array} \right.$$

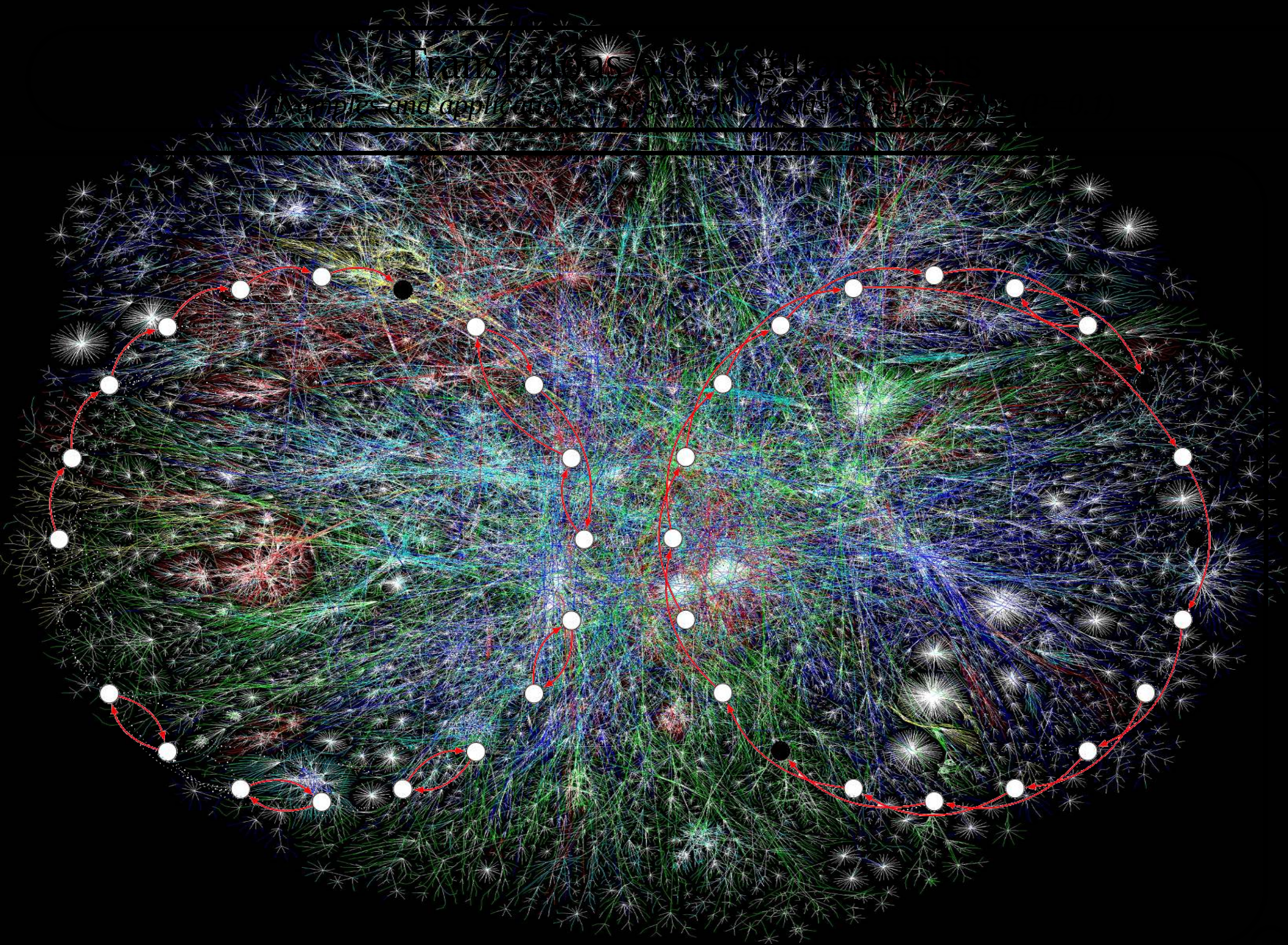
Restriction to edges

Injectivity

Avoid empty solution

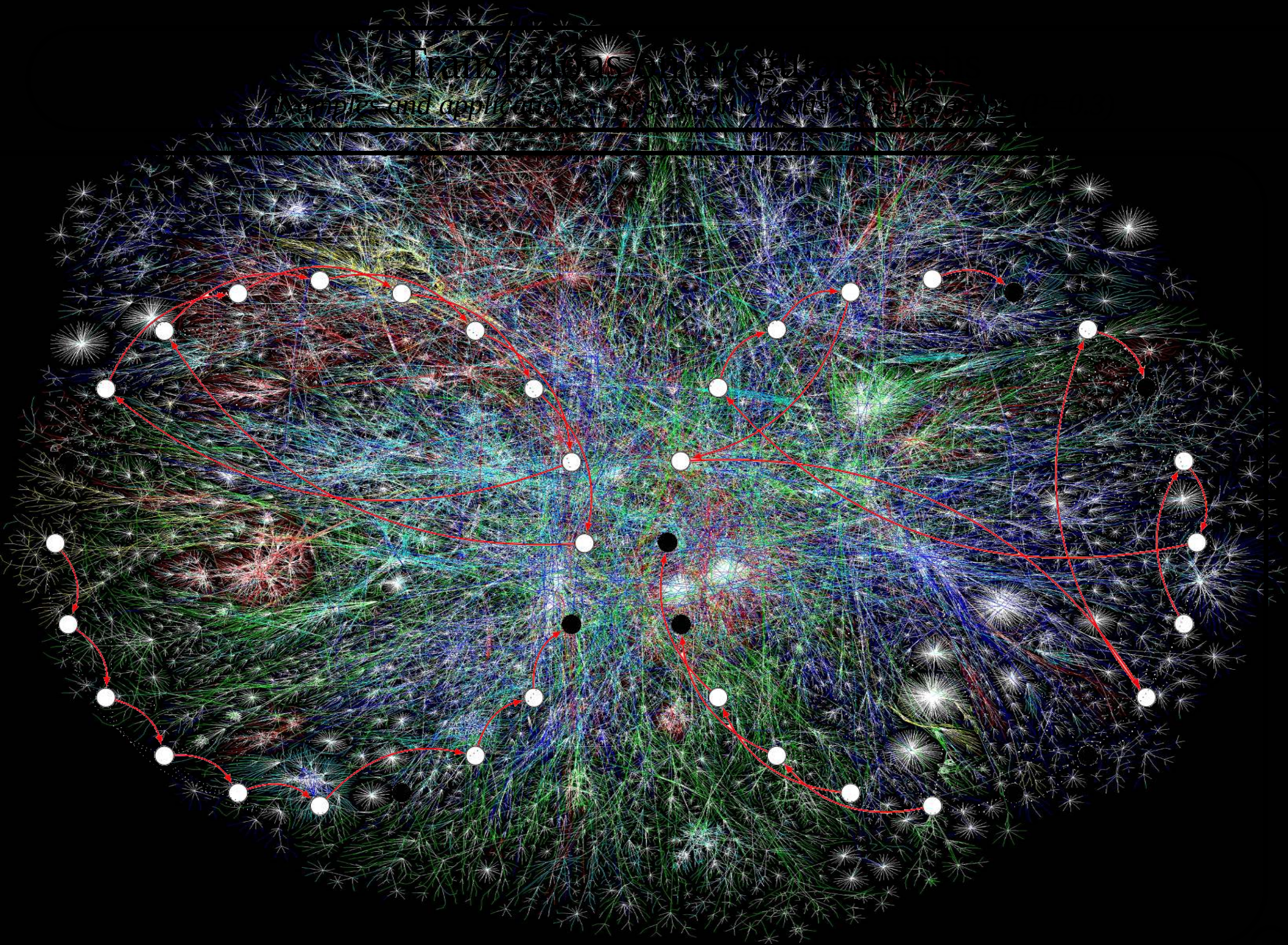
Transients

Models and applications in cosmology and astrophysics



Transients

Simple and applications to cosmological models

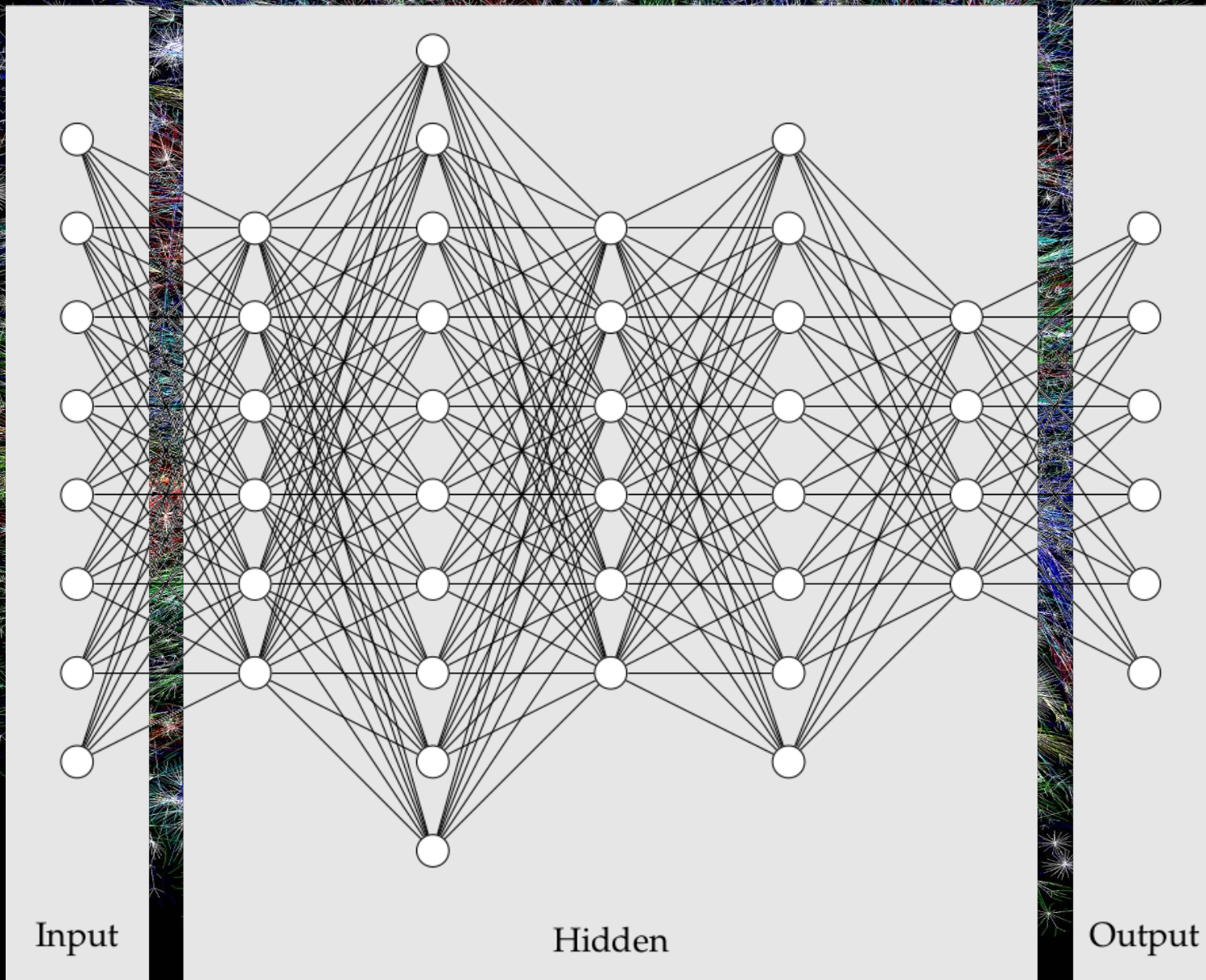


1.1.1

1.1.2

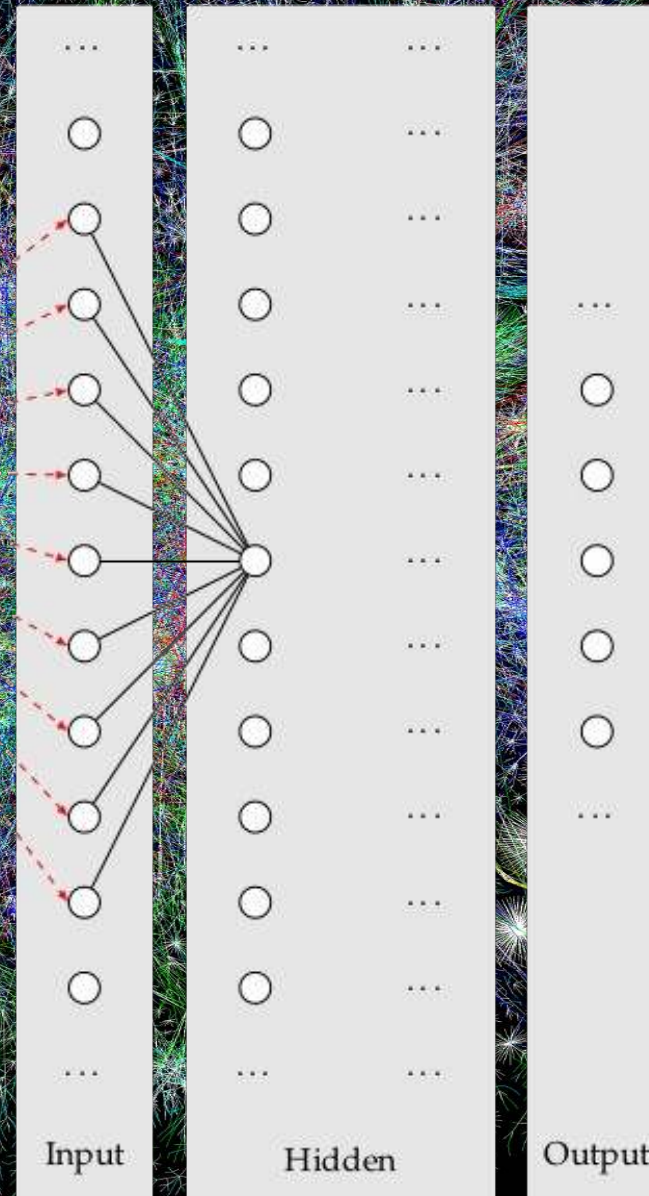
Example

1.1.3



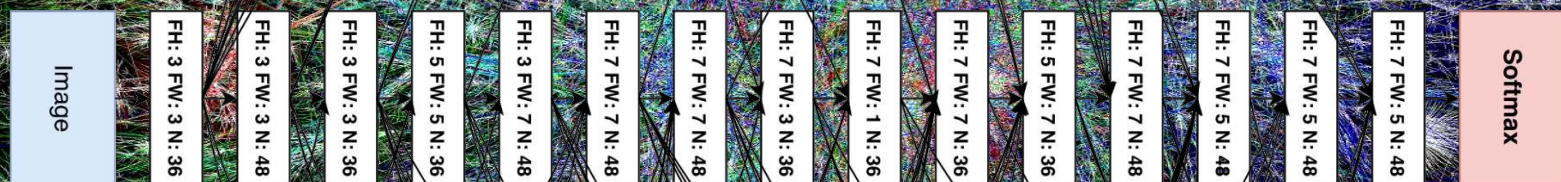
Transferring

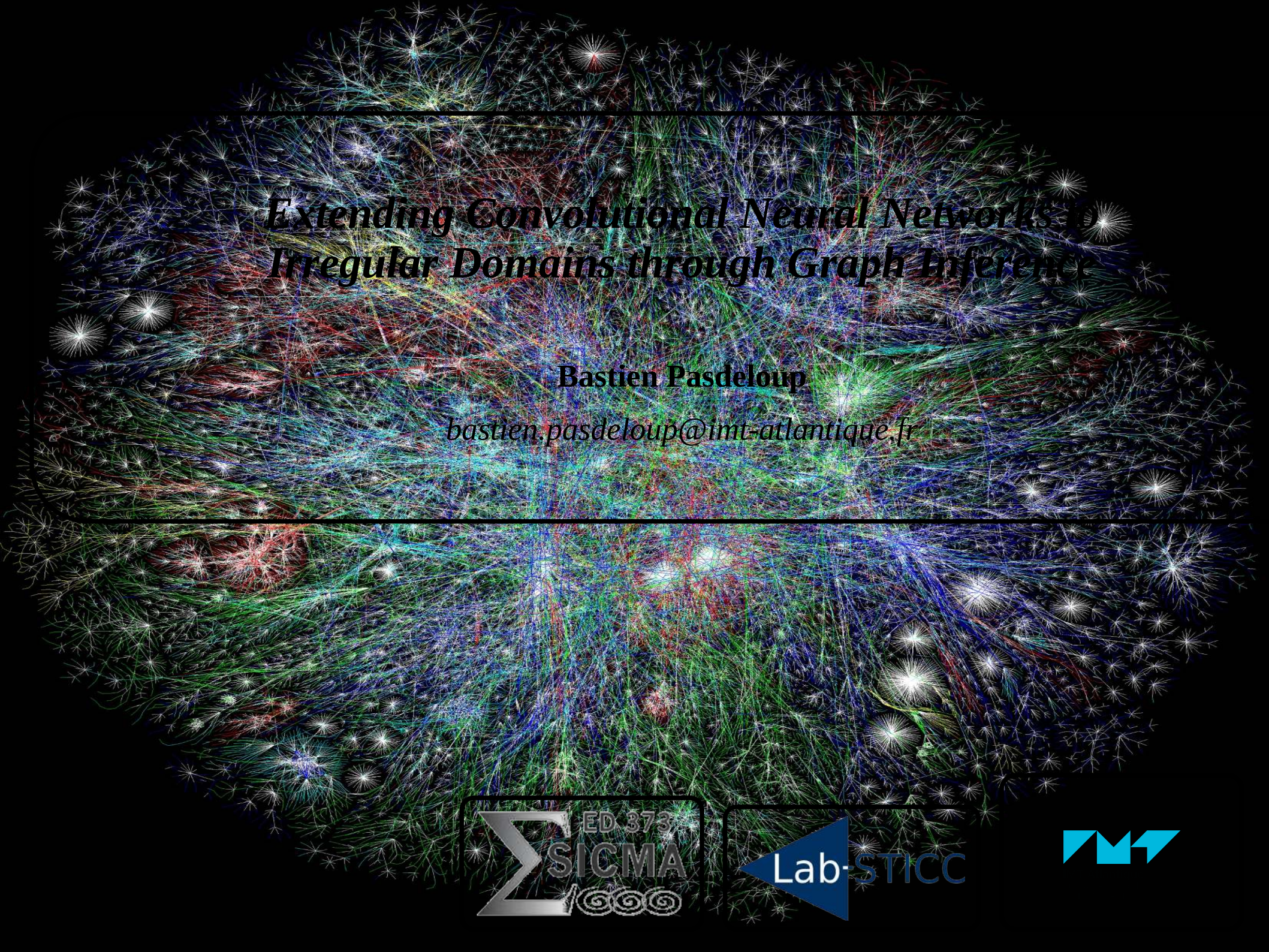
Examples and applications of deep learning



Transferrable

Examples and applications – Correlation of LUT





Extending Convolutional Neural Networks to Irregular Domains through Graph Inference

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